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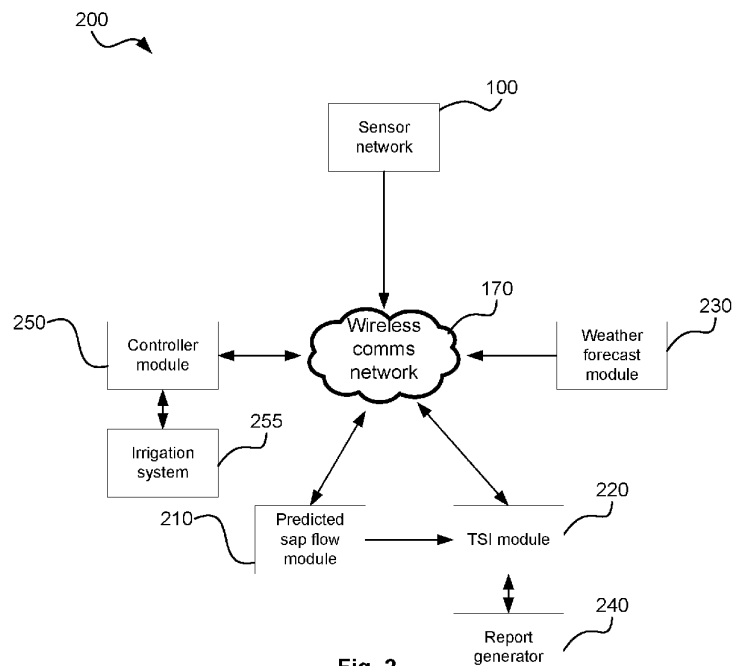


Fig. 2

(57) Abstract: Some embodiments relate to a method of detecting transpiration stress for at least one plant. The method comprises: receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant; receiving a plurality of predicted SFV data values associated to the at least one plant; and calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values. Some embodiments relate to a method of calculating predicted Sap Flow Velocity (SFV) for at least one plant. Some embodiments relate to a method of determining a correlation function for calculating predicted Sap Flow Velocity (SFV) for at least one plant. Some embodiments disclose related systems and computer programs.

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SYSTEMS AND METHODS FOR CROP MONITORING

FIELD

[0001] This invention relates generally to system and methods for automated crop monitoring using sap flow sensors and weather data. Crop monitoring has application, for example, in irrigation management.

BACKGROUND

[0002] Crop monitoring technology is utilised to automate drought sensing in crops, and more generally transpiration sensing, ensuring the precise scheduling of optimal water amounts at the right time for irrigation. Precise scheduling has the potential to maximize potential crop yields and optimize irrigation water usage, ensuring that crops do not receive too much or too little irrigation.

[0003] One form of crop monitoring technology includes the use of soil moisture probes. However, this method, especially in tree crops with varied root architecture, provides an overly simplistic way to predict plant water usage and stress levels. This crude soil-moisture monitoring approach offers only basic scheduling advice and often results in errors or inaccuracies, leading to potential over-irrigation of crops. Over-irrigation can adversely affect yields, particularly for sensitive crops like macadamias.

[0004] Another type of crop monitoring technology involves dendrometers, which are plant-based sensors designed to detect soil drought effects by measuring the daily swelling and shrinkage patterns of tree stems or developing fruit. However, the interpretations of signals from dendrometers are sometimes unreliable, which can lead to excessive irrigation when the soil is already saturated, even if other sensors indicate no drought stress.

[0005] It would be desirable to apply the right amount of water to closely match crop requirements, without exceeding them. Such application of water has the potential to lead to sustained yield increases in the medium and long term.

[0006] One or more embodiments of the present disclosure address or ameliorate at least one disadvantage or shortcoming of prior techniques, or at least provide a useful alternative thereto.

[0007] Any discussion of documents, acts, materials, devices, articles or the like which has been included in the present specification is not to be taken as an admission that any or all of these matters form part of the prior art base or were common general knowledge in the field relevant to the present disclosure as it existed before the priority date of each of the appended claims.

SUMMARY

[0008] Some embodiments of the present disclosure relate to a method of detecting transpiration stress for at least one plant. The method comprises: receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant; receiving a plurality of predicted SFV data values associated to the at least one plant; and calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

[0009] The term 'comprising' as used in this specification means 'consisting at least in part of'. When interpreting each statement in this specification that includes the term 'comprising', features other than that or those prefaced by the term may also be present. Related terms such as 'comprise' and 'comprises' are to be interpreted in the same manner.

[0010] Some embodiments of the present disclosure relate to a method of detecting transpiration stress for at least one plant. The method comprises: receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant; receiving a plurality of environmental data values associated to at least one environmental variable; determining a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values; and calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

[0011] The method may comprise calculating the TSI as a ratio of the predicted SFV data to the measured SFV data.

[0012] The method may further comprise: receiving at least one Soil Water Potential (SWP) data value; and assigning either a negative value or a positive value to the calculated TSI based at least partly on a comparison of the at least one SWP with a threshold.

[0013] The method may comprise assigning a negative value to the calculated TSI on determining that the at least one SWP is above the threshold. The method may comprise assigning a positive value to the calculated TSI on determining that the at least one SWP is below the threshold.

[0014] The threshold may be selected from within the range 0 to -30kPa.

[0015] At least some of the predicted SFV data values may be associated to respective timestamps. At least some of the measured Sap Flow Velocity (SFV) data values may be associated to respective timestamps. Calculating the Transpiration Stress Index (TSI) may comprise calculating the TSI at least partly from those measured SFV data values and those predicted SFV data values having timestamps within a bounded time interval.

[0016] The bounded time interval may comprise a time interval between 9am and 12pm. The bounded time interval may comprise a time interval between 9am and 2pm.

[0017] The method may further comprise generating an irrigation schedule recommendation at least partly from the calculated TSI.

[0018] The method may further comprise automatically controlling an irrigation system at least partly based on the calculated TSI. Automatically controlling the irrigation system may comprise controlling the irrigation system to increase irrigation when the TSI exceeds an upper TSI threshold and to cease irrigation when the TSI falls below a lower TSI threshold.

[0019] Some embodiments of the present disclosure relate to a method of calculating predicted Sap Flow Velocity (SFV) for at least one plant. The method comprises: receiving a plurality of environmental data values associated to at least one environmental variable; and determining a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at

least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

[0020] The at least one environmental variable may comprise Vapour Pressure Deficit (VPD), the correlation function configured to take as input VPD data values. Receiving the plurality of environmental data values may comprise receiving a plurality of VPD data values associated to VPD. Receiving the plurality of environmental data values may comprise: receiving a plurality of humidity data values; receiving a plurality of temperature data values; and deriving a plurality of VPD data values at least partly from the humidity data values and the temperature data values.

[0021] The at least one environmental variable may comprise Evapotranspiration (ET), the correlation function configured to take as input ET data values. Receiving the plurality of environmental data values may comprise receiving a plurality of ET data values associated to ET. Receiving the plurality of environmental data values may comprise: receiving a plurality of humidity data values; receiving a plurality of temperature data values; receiving a plurality of solar radiation data values; receiving a plurality of wind speed data values; and deriving a plurality of ET data values at least partly from the humidity data values, the temperature data values, the solar radiation data values, and the wind speed data values.

[0022] The at least one environmental variable may further comprise soil temperature, the correlation function configured to take as input VPD data values and soil temperature data values.

[0023] The at least one environmental variable further comprises solar radiation, the correlation function configured to take as input VPD data values and solar radiation data values.

[0024] At least some of the environmental data values may be associated to respective timestamps. Determining the predicted SFV may comprise the correlation function taking as input those environmental data values having timestamps within a bounded time interval.

[0025] The bounded time interval may comprise a time interval between 9am and 12pm. The bounded time interval may comprise a time interval between 9am and 2pm.

[0026] Some embodiments of the present disclosure relate to a method of determining a correlation function for calculating predicted Sap Flow Velocity (SFV) for at least one plant. The method comprises: receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant; receiving a plurality of environmental data values associated to at least one environmental variable; and determining a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

[0027] At least some of the measured Sap Flow Velocity (SFV) data values may be associated to respective timestamps. At least some of the environmental data values may be associated to respective timestamps. Determining the correlation function may comprise correlating those measured SFV data values and environmental data values having timestamps within a bounded time interval.

[0028] The bounded time interval may comprise a time interval between 9am and 12pm. The bounded time interval may comprise a time interval between 9am and 2pm.

[0029] Determining the correlation function may comprise correlating measured SFV data values and environmental data values measured in the absence of transpiration stress of the at least one plant.

[0030] The at least one environmental variable may comprise Vapour Pressure Deficit (VPD), the environmental data values comprising a plurality of VPD data values.

[0031] The at least one environmental variable may comprise Evapotranspiration (ET), the environmental data values comprising a plurality of ET data values.

[0032] The at least one environmental variable may comprise Vapour Pressure Deficit (VPD) and Evapotranspiration (ET), the environmental data values comprising a plurality of VPD data values and ET data values.

[0033] Some embodiments of the present disclosure relate to a transpiration stress detection system for at least one plant. The system comprises: a Transpiration Stress Index (TSI) module configured to: receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant, receive a plurality of predicted SFV data values associated

to the at least one plant, and calculate a TSI at least partly from the measured SFV data values and the predicted SFV data values; and an output device configured to output the calculated TSI.

[0034] Some embodiments of the present disclosure relate to a transpiration stress detection system for at least one plant. The system comprises: a predicted sap flow module configured to: receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant, receive a plurality of environmental data values associated to at least one environmental variable, and determine a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate the measured SFV data values associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values; a Transpiration Stress Index (TSI) module configured to: receive the plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant, receive the plurality of predicted SFV data values determined by the predicted sap flow module, and calculate a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values; and an output device configured to output the calculated TSI.

[0035] The TSI module may be further configured to calculate the TSI as a ratio of the predicted SFV data to the measured SFV data.

[0036] The TSI module may be further configured to: receive at least one Soil Water Potential (SWP) data value; and assign either a negative value or a positive value to the calculated TSI based at least partly on a comparison of the at least one SWP with a threshold.

[0037] The TSI module may be configured to assign a negative value to the calculated TSI on determining that the at least one SWP is above the threshold. The TSI module may be configured to assign a positive value to the calculated TSI on determining that the at least one SWP is below the threshold.

[0038] The threshold may be selected from within the range 0 to -30kPa.

[0039] At least some of the predicted SFV data values are associated to respective timestamps. At least some of the measured Sap Flow Velocity (SFV) data values are associated to respective timestamps. Calculating the Transpiration Stress Index (TSI)

comprises calculating the TSI at least partly from those measured SFV data values and those predicted SFV data values having timestamps within a bounded time interval.

[0040] The bounded time interval may comprise a time interval between 9am and 12pm. The bounded time interval may comprise a time interval between 9am and 2pm.

[0041] The system may further comprise a report generator configured to generate an irrigation schedule recommendation at least partly from the calculated TSI.

[0042] The system may further comprise a controller module configured to automatically control an irrigation system at least partly based on the calculated TSI. Automatically controlling the irrigation system may comprise controlling the irrigation system to increase irrigation when the TSI exceeds an upper TSI threshold and to cease irrigation when the TSI falls below a lower TSI threshold.

[0043] Some embodiments of the present disclosure relate to a predicted Sap Flow Velocity (SFV) calculation system for at least one plant. The system comprises: a predicted sap flow module configured to: receive a plurality of environmental data values associated to at least one environmental variable; and determine a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

[0044] The at least one environmental variable may comprise Vapour Pressure Deficit (VPD), the correlation function configured to take as input VPD data values. Receiving the plurality of environmental data values may comprise receiving a plurality of VPD data values associated to VPD. Receiving the plurality of environmental data values may comprise: receiving a plurality of humidity data values; receiving a plurality of temperature data values; and deriving a plurality of VPD data values at least partly from the humidity data values and the temperature data values.

[0045] The at least one environmental variable may comprise Evapotranspiration (ET), the correlation function configured to take as input ET data values. Receiving the plurality of environmental data values may comprise receiving a plurality of ET data values associated to ET. Receiving the plurality of environmental data values may comprise: receiving a plurality of humidity data values; receiving a plurality of temperature data values; receiving a plurality of

solar radiation data values; receiving a plurality of wind speed data values; and deriving a plurality of ET data values at least partly from the humidity data values, the temperature data values, the solar radiation data values, and the wind speed data values.

[0046] The at least one environmental variable may further comprise soil temperature, the correlation function configured to take as input VPD data values and soil temperature data values.

[0047] The at least one environmental variable may further comprise solar radiation, the correlation function configured to take as input VPD data values and solar radiation data values.

[0048] At least some of the environmental data values may be associated to respective timestamps. Determining the predicted SFV may comprise the correlation function taking as input those environmental data values having timestamps within a bounded time interval.

[0049] The bounded time interval may comprise a time interval between 9am and 12pm. The bounded time interval may comprise a time interval between 9am and 2pm.

[0050] Some embodiments of the present disclosure relate to a correlation function determination system for calculating predicted Sap Flow Velocity (SFV) for at least one plant. The system comprises: a predicted sap flow module configured to: receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant; receive a plurality of environmental data values associated to at least one environmental variable; and determine a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

[0051] At least some of the measured Sap Flow Velocity (SFV) data values may be associated to respective timestamps. At least some of the environmental data values may be associated to respective timestamps. Determining the correlation function may comprise correlating those measured SFV data values and environmental data values having timestamps within a bounded time interval.

[0052] The bounded time interval may comprise a time interval between 9am and 12pm. The bounded time interval comprises a time interval between 9am and 2pm.

[0053] Determining the correlation function may comprise correlating measured SFV data values and environmental data values measured in the absence of transpiration stress of the at least one plant.

[0054] The at least one environmental variable may comprise Vapour Pressure Deficit (VPD), the environmental data values comprising a plurality of VPD data values.

[0055] The at least one environmental variable may comprise Evapotranspiration (ET), the environmental data values comprising a plurality of ET data values.

[0056] The at least one environmental variable may comprise Vapour Pressure Deficit (VPD) and Evapotranspiration (ET), the environmental data values comprising a plurality of VPD data values and ET data values.

[0057] Some embodiments of the present disclosure relate to a transpiration stress detection system for at least one plant. The system comprises: a storage device; and a processor programmed to: receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant; receive a plurality of predicted SFV data values associated to the at least one plant; and calculate a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

[0058] Some embodiments of the present disclosure relate to a transpiration stress detection system for at least one plant, the system comprising: a storage device; and a processor programmed to: receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant, receive a plurality of environmental data values associated to at least one environmental variable, determine a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values, and calculate a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

[0059] Some embodiments of the present disclosure relate to a predicted Sap Flow Velocity (SFV) calculation system for at least one plant, the system comprising: a storage device; and a processor programmed to: receive a plurality of environmental data values associated to at least one environmental variable, and determine a predicted SFV for the at least one plant

using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

[0060] Some embodiments of the present disclosure relate to a correlation function determination system for calculating predicted Sap Flow Velocity (SFV) for at least one plant, the system comprising: a storage device; and a processor programmed to: receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant, receive a plurality of environmental data values associated to at least one environmental variable, and determine a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

[0061] Some embodiments of the present disclosure relate to a computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of detecting transpiration stress for at least one plant, the method comprising: receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant; receiving a plurality of predicted SFV data values associated to the at least one plant; and calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

[0062] Some embodiments of the present disclosure relate to a computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of detecting transpiration stress for at least one plant, the method comprising: receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant; receiving a plurality of environmental data values associated to at least one environmental variable; determining a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values; and calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

[0063] Some embodiments of the present disclosure relate to a computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of calculating predicted Sap Flow Velocity (SFV) for

at least one plant, the method comprising: receiving a plurality of environmental data values associated to at least one environmental variable; and determining a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

[0064] Some embodiments of the present disclosure relate to a computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of determining a correlation function for calculating predicted Sap Flow Velocity (SFV) for at least one plant, the method comprising: receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant; receiving a plurality of environmental data values associated to at least one environmental variable; and determining a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

[0065] Some embodiments of the present disclosure relate to a method of detecting soil drought stress for a type of plant, the method comprising: calculating a drought stress calibration curve for the type of plant comprising: installing sap flow sensors in a plurality of plants of the type, obtaining sap flow velocity data from the sap flow sensors, measuring temperature and humidity data on-site, and calculating the drought stress calibration curve using the sap flow velocity data and the temperature and humidity data; and detecting soil drought stress for the type of plant by: obtaining live sap flow velocity data from the sap flow sensors, calculating a predicted sap flow using live temperature and humidity (VPD) data and the drought stress calibration curve, and detecting a soil drought stress index as the residual between the live sap flow velocity data and the predicted sap flow.

[0066] A drought stress calibration curve may be calculated for each plant. The method may further comprise calculating a drought stress calibration curve from each drought stress calibration curve.

[0067] Calculating the drought stress calibration curve may be performed during a well-watered period.

[0068] The method may further comprise automatically controlling an irrigation system according to the soil drought stress index.

[0069] Automatically controlling the irrigation system may comprise controlling the irrigation system to increase irrigation when the soil drought stress index falls beneath a threshold and to cease increasing irrigation when the soil drought stress index exceeds the threshold.

[0070] The techniques described herein have the potential to allow for the fine-tuning of irrigation schedules to improve water-use efficiency and maximize yields.

[0071] In an embodiment, the plant is an angiosperm. In an embodiment, the plant is a dicotyledonous plant.

[0072] In an embodiment, the plant has a stem which is at least 8mm in diameter. In an embodiment, the plant has a stem which is at least 25mm in diameter. In an embodiment, the plant has a stem which is at least 50mm in diameter. In an embodiment, the plant has a stem which is at least 100mm in diameter. In an embodiment, the plant has a stem which is at least 200mm in diameter. In an embodiment, the plant has a stem which is at least 300mm in diameter. In an embodiment, the plant has a stem which is at least 400mm in diameter. In an embodiment, the plant has a stem which is at least 500mm in diameter. In an embodiment, the plant has a stem which is at least 600mm in diameter. In an embodiment, the plant has a stem which is at least 700mm in diameter. In an embodiment, the plant has a stem which is at least 800mm in diameter. In an embodiment, the plant has a stem which is at least 900mm in diameter. In an embodiment, the plant has a stem which is at least 1,000mm in diameter.

[0073] In an embodiment, the plant has a stem which is between 8mm and 1,000mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 900mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 800mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 700mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 600mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 500mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 400mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 300mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 200mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 100mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 50mm in diameter. In an embodiment, the plant has a stem which is between 8mm and 25mm in diameter.

[0074] As used herein, phrases referring to a diameter of a stem is referring to at least one portion of the stem having the defined diameter.

[0075] In an embodiment, the plant produces nuts, fruit such as stone fruits, vegetables or fibre.

[0076] Non-limiting examples of plants which can be analysed using the methods of the invention include, but are not limited to, a fruit crop such as apples, apricots, cherries, nectarines, peaches, pears, plums, prunes, cherry, quince, fig, nuts (e.g., chestnuts, brazil nuts, cashews, pecans, pistachios, hazelnuts, pistachios, peanuts, almonds, walnuts, macadamia nuts, and the like), citrus (e.g., clementine, kumquat, orange, grapefruit, tangerine, mandarin, lemon, lime, and the like), blueberries, black raspberries, boysenberries, cranberries, currants, gooseberries, loganberries, raspberries, strawberries, blackberries, grapes (wine and table), avocados, bananas, kiwi, persimmons, pomegranate, pineapple, tropical fruits, pomes, mango, papaya, and lychee, sorghum, tobacco, kapok, a leguminous plant (beans (e.g., green and dried), lentils, peas, soybeans), an oil plant (e.g. olive, sunflower, coconut, castor oil plant, cocoa bean, oil palm), a fiber plant (e.g. cotton, hemp), Cannabis (e.g., Cannabis saliva, Cannabis indica, and Cannabis ruderalis), lauraceae (cinnamon, camphor), coffee bean, sugar cane, tea, natural rubber plants, an ornamental plant (e.g., roses), trees such as forest trees (broad-leaved trees and evergreens, such as conifers; e.g., elm, ash, oak, maple, fir, spruce, cedar, pine, birch, cypress, eucalyptus, willow), as well as shrubs and other nursery stock.

[0077] In an embodiment, the stem is a trunk of a woody plant. In an embodiment, the stem is a trunk of a non-woody plant. By "woody plants" is meant plants that exhibit secondary growth originating in the cambium.

[0078] The invention in one aspect comprises several steps. The relation of one or more of such steps with respect to each of the others, the apparatus embodying features of construction, and combinations of elements and arrangement of parts that are adapted to affect such steps, are all exemplified in the following detailed disclosure.

[0079] To those skilled in the art to which the invention relates, many changes in construction and widely differing embodiments and applications of the invention will suggest themselves without departing from the scope of the invention as defined in the appended claims. The disclosures and the descriptions herein are purely illustrative and are not intended to be in any

sense limiting. Where specific integers are mentioned herein which have known equivalents in the art to which this invention relates, such known equivalents are deemed to be incorporated herein as if individually set forth.

[0080] As used herein, '(s)' following a noun means the plural and/or singular forms of the noun.

[0081] As used herein, the term 'and/or' means 'and' or 'or' or both.

[0082] In the description in this specification reference may be made to subject matter which is not within the scope of the appended claims. That subject matter should be readily identifiable by a person skilled in the art and may assist in putting into practice the invention as defined in the presently appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0083] Preferred forms of the systems and methods for crop monitoring, including irrigation management, will now be described by way of example only with reference to the accompanying figures in which:

[0084] Figure 1 shows an example of a sensor network configured to obtain data for crop monitoring.

[0085] Figure 2 shows an example of an irrigation management system.

[0086] Figure 3 shows a method of calculating predicted Sap Flow Velocity (SFV).

[0087] Figure 4 shows a method of determining a Transpiration Stress Index (TSI) for a type of plant using live sap flow sensor data and predicted Sap Flow Velocity (SFV).

[0088] Figures 5 to 8 show examples of application of the system of figure 2 to a cotton plant.

[0089] Figures 9 to 12 show examples of application of the system of figure 2 to a macadamia tree. The figures illustrate data obtained from a first sensor.

[0090] Figures 13 to 16 show examples of application of the system of figure 2 to a macadamia tree. The figures illustrate data obtained from a second sensor.

[0091] Figure 17 shows the influence that soil temperature may have on Sap Flow Velocity (SFV).

[0092] Figure 18 shows an example of a recommended irrigation schedule generated by the irrigation management system of figure 2.

[0093] Figure 19 illustrates an example of a computing device configured to implement components of the irrigation management system of figure 2.

DETAILED DESCRIPTION

[0094] **Figure 1** shows an example of a sensor network 100. Sensor network 100 may be configured to obtain data about a crop for the purposes of crop monitoring. Examples of the data obtained about crops are described below.

[0095] Shown in figure 1 are examples of plants, for example plant 105 and plant 110. Plants 105 and 110 may, for example, be positioned within a crop area and form part of an irrigated crop.

[0096] Plant 105 may have an associated sap flow sensor 115 configured to measure sap flow within plant 105. Sap flow sensor 115 may take various forms. For example, sap flow sensor 115 may be configured to be at least partially located within a xylem of plant 105. Sap flow sensor 115 may be configured to encircle a stem of plant 105 without penetrating the stem. Sap flow sensor 115 may be configured to be positioned in close proximity to plant 105 without contacting the stem.

[0097] Plant 110 may have an associated sap flow sensor 120 configured to measure sap flow within plant 110. Sap flow sensor 120 may be configured similarly to sap flow sensor 115. For example, sap flow sensor 120 may be located within, around, or in close proximity to a stem of plant 110.

[0098] Typically, sap flow sensor 115 and/or sap flow sensor 120 may be configured to measure sap flow velocity at two points in the stem (~1-1.5 and 2-2.5cm in from the base of the needles) of plant 105 and plant 110 respectively on 8–15 minute intervals and directly measure instantaneous crop water-use or transpiration. The intervals may be 8-10 minutes or larger. The data obtained from sap flow sensor 115 and/or sensor 120 may be associated to respective timestamps. The timestamps represent a time or day at which sap flow sensor 115

or sap flow sensor 120 obtained respective data values representing Sap Flow Velocity (SPV) associated to plant 105 or plant 110.

[0099] Data representing measured or observed Sap Flow Velocity (SPV) obtained by sap flow sensor 115 and/or sap flow sensor 120 may be transmitted to at least one data logger 170. In one example, sap flow sensor 115 and sap flow sensor 120 are connected physically to respective data loggers. In this case, there are multiple pairs of sap flow sensors and data loggers. In another example, multiple sap flow sensors are in wireless communication with data logger 170. In this case multiple sap flow sensors are in wireless communication with a single data logger.

[0100] Sensor network 100 may further include a humidity sensor 140 and/or temperature sensor 145. It will be appreciated that humidity sensor 140 and temperature sensor 145 may be configured as two separate sensors. It will also be appreciated that a single sensor could be used that measures both humidity and temperature.

[0101] Sensor network 100 may include additional environmental sensors. These environmental sensors include, for example, soil moisture sensor 150, soil temperature sensor 155, solar radiation sensor 160, and/or wind speed sensor 165. It will be appreciated that some or all of these sensors may be combined into one or more multi-purpose sensors. These additional environmental sensors may be connected physically or wirelessly to data logger(s) 170.

[0102] The data obtained from the environmental sensors 140, 145, 150, 155, 160 and/or 165 may be associated to respective timestamps. The timestamps represent a time or day at which the environmental sensors obtained respective data values representing respective environmental conditions.

[0103] Sensor network 100 may further include at least one data logger 150. In this example, humidity sensor 140 and temperature sensor 145 are physically or wirelessly connected to data logger 150. In another example, humidity sensor 140 and temperature sensor 145 are physically or wirelessly connected to separate data loggers.

[0104] Data logger(s) 170 and data logger(s) 150 may be physically or wirelessly connected to a data communications module 180. Data communications module 180 may be connected to wireless communications network 190. Data obtained from one or more of the sensors

forming part of sensor network 100 may be transmitted over wireless communications network 190.

[0105] **Figure 2** shows an example of an irrigation management system 200. System 200 includes a sensor network. One example of a sensor network is sensor network 100 from figure 1. Data obtained from sensor network 100 may be transmitted to other components of system 200 via a wireless communications network, for example network 190 from figure 1.

[0106] System 200 includes a predicted sap flow module 205. Module 205 may be configured to receive, over network 190, humidity and temperature values from sensor network 100. For example, the humidity and temperature values may be obtained from humidity sensor 140 and temperature sensor 145 from figure 1 respectively.

[0107] Crop transpiration is significantly affected by temperature and humidity. These two values may be combined in one variable, Vapour Pressure Deficit (VPD), by calculating the difference between the vapour pressure at 100% (saturated) humidity for the given temperature and an estimate of live vapour pressure at the given humidity.

[0108] Vapour Pressure Deficit (VPD) is a measure of the major proportion of evaporative pressure that determines the rate of evaporation of water from the stomata on the leaves and other atmospherically exposed organs of the plants of a crop.

[0109] Evapotranspiration (ET) is broadly defined as a combination of evaporation and transpiration. Evaporation is the movement of water directly to the air from sources such as the soil and water bodies. ET can be affected by factors such as heat, humidity, solar radiation and wind speed. Transpiration is the movement of water through the stomata of plants from the crop. An ET rate may be measured for example in millimetres of water in a set unit of time.

[0110] Predicted sap flow module 205 may be configured to determine a calibration curve specific to plants of a crop, for example plant 105 and/or plant 110. This calibration curve may include a relationship between Sap Flow Velocity (SPV) and Vapour Pressure Deficit (VPD). The calibration curve may include a relationship between SPV and ET rate.

[0111] Predicted Sap Flow Velocity module 205 may be configured to determine a predicted SPV for plant 105 and/or plant 110 based at least partly on a calibration curve determined by module 205.

[0112] System 100 may include a Transpiration Stress Index (TSI) module 220.

[0113] Generally speaking, TSI module 220 receives data from sensor network 100 representing Sap Flow Velocity (SPV) associated to, for example, plant 105 and plant 110. TSI module 220 may also receive data from predicted sap flow module 210 representing predicted Sap Flow Velocity (SPV) associated to plant 105 and plant 110.

[0114] TSI module 220 may be configured to calculate a Transpiration Stress Index (TSI) based at least partly from predicted SPV and measured SPV for plant 105 and/or plant 110. For example, TSI may be calculated as a ratio of predicted SPV to measured SPV as follows:

$$TSI = \frac{\text{predicted SPV}}{\text{measured SPV}}$$

[0115] TSI module 220 may be configured to calculate the TSI based at least partly from predicted SPV, measured SPV, and soil moisture data representing Soil Water Potential (SWP) obtained, for example, from soil moisture sensor 150.

[0116] TSI module 220 may be configured to receive SWP data and assign either a negative value or a positive value to the calculated TSI based at least partly on a comparison of the SWP data with a threshold.

[0117] For example, a negative value may be assigned to the calculated TSI if the SWP is above the threshold. In situations where the SWP is above the threshold, the TSI module 220 may determine that plant 105 and/or plant 110 is in a waterlogged state. Assigning a negative value to the calculated TSI indicates that plant 105 and/or plant 110 does not require any irrigation.

[0118] A positive value may be assigned to the calculated TSI if the SWP is below the threshold. In situations where the SWP is below the threshold, the TSI module 220 may determine that plant 105 and/or plant 110 is in a drought state. Assigning a positive value to the calculated TSI indicates that plant 105 and/or plant 110 requires irrigation.

[0119] The threshold may be selected from within a range of 0 to -30kPa. The threshold may be selected from within a range of 0 to -12kPa. The threshold may be selected from within a range of -8kPa to -12kPa.

[0120] In another example, if SWP is between 0 and 10kPa then the TSI calculated from the above equation may be assigned a negative value. On the other hand, if SWP is less than -10kPa then the TSI calculated from the above equation may be assigned a positive value.

[0121] The following table shows indicative TSI values calculated from predicted SPV, measured SPV, and soil moisture data and their relationship to transpiration stress descriptions.

TSI value	Transpiration stress description
$TSI > 2.0$	Drought high stress
$1.5 < TSI < 2.0$	Drought low stress
$-1.5 < TSI < 1.5$	No transpiration stress
$-2.0 < TSI < -1.5$	Waterlogged low stress
$TSI < -2.0$	Waterlogged high stress

[0122] It will be appreciated that the boundary values defining transpiration stress categories can be varied. For example, the values ± 1.5 and ± 2.0 are shown for the purposes of illustration and may be varied. It will also be appreciated that the category boundary values ± 1.5 and ± 2.0 may be included in either one of the categories for which the TSI represents a boundary. For example, TSI module 220 may be configured to determine a category of waterlogged low stress from one of the following ranges of TSI values:

$-2.0 \leq TSI < -1.5$; or

$-2.0 \leq TSI \leq -1.5$; or

$-2.0 < TSI \leq -1.5$; or

$-2.0 < TSI < -1.5$.

[0123] TSI module 220 may be configured to determine a TSI periodically, for example at intervals of a time value between approximately 8 minutes and approximately 15 minutes. It will be appreciated that the intervals can range from every few minutes to once per day or longer. The size of the time intervals depends on system requirements and sensor configuration.

[0124] Where TSI module 220 is configured to determine a TSI periodically, the output may comprise a plurality of TSI data values. At least some of the TSI data values may be associated to respective timestamps. The timestamps may indicate, for example, a time and/or date at which measured SFV data values in relation to plant 105 and/or plant 110 were obtained from sensor network 100.

[0125] TSI module 220 may be configured to select at least some of the TSI data values having timestamps that are within a bounded time interval. The bounded time interval may comprise a time interval between 9am and 12pm. The bounded time interval may comprise a time interval between 9am and 10am. The bounded time interval may comprise a time interval between 9am and 2pm. The bounded time interval may comprise a time interval between 5pm and 8pm. The bounded time interval may comprise a time interval between 7pm and 10pm.

[0126] TSI module 220 may be configured to calculate a single TSI data value that is representative of the selection of TSI data values. For example, TSI module 220 may calculate an average or mean of the selection of TSI data values in order to calculate the single TSI data value. This single TSI data value may be used, for example to indicate transpiration stress for plant 105 and/or plant 110 on a selected day.

[0127] System 200 may include a weather forecast module 230 which may be configured to transmit data representing atmospheric conditions via network 170 to other components of system 200.

[0128] System 100 may include a report generator 240 configured to determine a recommended irrigation schedule based at least partly on data from TSI module 220 and weather forecast module 230.

[0129] System 100 may include a controller module 250 connected to an irrigation system 255. The controller module 250 may be configured to receive data from TSI module 220 via network 170. Controller module 250 may be configured to interpret the received data, including

for automated drought detection and automation. In the embodiment shown, controller module 250 may automatically control sprinkler valves of an irrigation system 255 based at least partly on data received from TSI module 220.

[0130] **Figure 3** shows a method 300 of calculating predicted Sap Flow Velocity (SFV) using, for example, predicted sap flow module 210. Method 300 calculates a calibration curve for plant 105 and/or plant 110.

[0131] Crops tend to regulate their water-use based on physiological adaptations evolved to survive in their original wild natural environment. Different crops have evolved rather different strategies for water-use responses to the environment which are often highly species-specific and usually based around different stomatal control mechanisms. The calibration curve seeks to determine a relationship between measured transpiration (sap flow) and VPD and/or ET, especially as conditions move from low VPD or ET conditions to high VPD or ET conditions.

[0132] Method 30 may comprise installing 305 sap flow sensors, for example sap flow sensor 115 and sap flow sensor 120, in relation to a plurality of plants, for example plant 105 and plant 110. As described above, sap flow sensors may be located within, around, or in close proximity to a stem of plant. The method further includes obtaining 310 Sap Flow Velocity (SPV) from sap flow sensor 115 and sap flow sensor 120.

[0133] The method may include obtaining 315 environmental data from environmental sensors. Examples of environmental sensors include humidity sensor 140, temperature sensor 145, soil moisture sensor 150, soil temperature sensor 155, solar radiation sensor 160, and/or wind speed sensor 165.

[0134] The method may include calibrating 320 measured Sap Flow Velocity (SFV) with environmental data to determine one or more calibration curves. SFV may be calibrated, for example, with Vapour Pressure Deficit (VPD) and/or evapotranspiration (ETs) values. Calibration may involve, for example, determining a correlation function for calculating predicted SFV. The correlation function may be configured to correlate measured SFV for plant 105 and/or plant 110 and at least one environmental variable. Examples of environmental variables include VPD and ET.

[0135] Predicted sap flow module 210 may be configured to receive a plurality of measured SFV data values and a plurality of environmental data values. At least some of the measured

SFV data values and environmental data values may be associated to respective timestamps. The timestamps may indicate, for example, a time and/or date at which measured SFV data values in relation to plant 105 and/or plant 110 and environmental data values were obtained from sensor network 100.

[0136] Predicted sap flow module 210 may be configured to select at least some of the SFV data values and/or environmental data values having timestamps that are within a bounded time interval. The bounded time interval may comprise a time interval between 9am and 12pm. The bounded time interval may comprise a time interval between 9am and 10am. The bounded time interval may comprise a time interval between 9am and 2pm. The bounded time interval may comprise a time interval between 5pm and 8pm. The bounded time interval may comprise a time interval between 7pm and 10pm.

[0137] Predicted sap flow module 210 may be configured to determine the correlation function based on the selected SFV data values and/or environmental data values.

[0138] The method may include determining 325 predicted SFV data which are then used by other components of system 200, for example TSI module 220. For example, the correlation function may take as input an environmental variable such as VPD or ET. The correlation function may output a predicted SFV given an input VPD or ET.

[0139] For the purposes of determining predicted SFV data, predicted sap flow module 210 may be configured to receive a plurality of environmental data values associated to at least one environmental variable. Examples of environmental data variables include VPD and ET.

[0140] As described above, at least some of the environmental data values may be associated to respective timestamps. The timestamps may indicate, for example, a time and/or date at which environmental data values were obtained from sensor network 100.

[0141] Predicted sap flow module 210 may be configured to select at least some of the environmental data values having timestamps that are within a bounded time interval. The bounded time interval may comprise a time interval between 9am and 12pm. The bounded time interval may comprise a time interval between 9am and 10am. The bounded time interval may comprise a time interval between 9am and 2pm. The bounded time interval may comprise a time interval between 5pm and 8pm. The bounded time interval may comprise a time interval between 7pm and 10pm.

[0142] Predicted sap flow module 210 may be configured to calculate a single predicted SFV data value that is representative of the selection of environmental data values. For example, predicted sap flow module 210 may calculate an average or mean of the selection of predicted SFV data values in order to calculate the single predicted SFV data value. This single predicted SFV data value may be used, for example to indicate a predicted SFV for plant 105 and/or plant 110 on a selected day.

[0143] **Figure 4** shows a method 400 of detecting transpiration stress for plant 105 and plant 110 which includes obtaining 405 live sap flow sensor data from sap flow sensor 115 and sap flow sensor 120.

[0144] The method includes receiving 410 predicted Sap Flow Velocity (SFV) data, for example from predicted sap flow module 210. A Transpiration Stress Index (TSI) is determined 415 based at least partly from the live sap flow sensor data and the predicted Sap Flow Velocity data.

[0145] The calculated TSI is then output 420. Report generator 240, for example may use the calculated TSI to generate irrigation schedule recommendations. Controller module 250 may use the calculated TSI to control sprinkler valves of irrigation system 255.

[0146] The calibration 320 of figure 3 of SFV and environmental data in well-watered situations (where soil moisture deficit does not limit transpiration) allows for the determining 415 of a Transpiration Stress Index (TSI) of figure 3 for detecting transpiration stress for the type of plant by identifying soil drought effects on trees which exhibit sap flow or water-use patterns that do not behave as expected for a well-watered tree.

[0147] Specifically, whereas sap flow patterns from a well-watered tree may be expected to predictably correlate with atmospheric VPD and/or ET, during mild to moderate soil drought, the measured sap flow behaviour begins to slightly de-couple from the standard response to atmospheric demands due to low soil water potential effects on the stomata.

[0148] Figures 5 to 8 show examples of application of system 200 to a cotton plant. Use of weather forecast module 230 from figure 2 is not displayed. It will be appreciated that the examples may be displayed differently when data from weather forecast module 230 is obtained.

[0149] **Figure 5** shows the relationship between Sap Flow Velocity (SFV) and Vapour Pressure Deficit (VPD) for one cotton plant during a four-day calibration period from 8th February 2024 to 11th February 2024, when the cotton plant was not subjected to transpiration stress. The data shows a strong positive correlation between SFV and VPD under these conditions, suggesting that Sap Flow Velocity (SFV) increases proportionally with Vapour Pressure Deficit (VPD) in the absence of transpiration stress.

[0150] **Figure 6** shows the correlation curve between Sap Flow Velocity (SFV) and Vapour Pressure Deficit (VPD) in a cotton plant, using data between collected between 9am and 12 pm. The equation derived from this correlation curve is critical for modelling the predicted Sap Flow Velocity (SFV). It is noted that different crops may necessitate distinct calibration periods, and adjustments should be made accordingly.

[0151] **Figure 7** demonstrates an observation that the predicted Sap Flow Velocity (SFV) for morning periods closely aligns with the actual Sap Flow Velocity (SFV) when the cotton plant was not experiencing transpiration stress.

[0152] **Figure 8** illustrates the average predicted of Sap Flow Velocity (SFV) for the morning period (9 am to 12 pm) over a 46 day window from 8th February to 24th March. Prolonged transpiration stress was observed from 7th March, evidenced by the discrepancy between the predicted and actual Sap Flow Velocities (SFV). Soil water potential decreases significantly during this transpiration stress period.

[0153] Figures 9 to 12 show examples of application of system 200 to macadamia trees. The figures illustrate data obtained from a first sensor. Use of weather forecast module 230 from figure 2 is not displayed. It will be appreciated that the examples may be displayed differently when data from weather forecast module 230 is obtained.

[0154] **Figure 9** illustrates the relationship between Sap Flow Velocity (SFV) and Vapour Pressure Deficit (VPD) of one macadamia tree during a four-day calibration period from 6th June 2024 to 9th June 2024, when the macadamia tree was not subjected to transpiration stress. The data indicates a strong positive correlation between SFV and VPD under these conditions, suggesting that Sap Flow Velocity increases proportionally with Vapour Pressure Deficit (VPD) in the absence of transpiration stress.

[0155] **Figure 10** depicts the correlation curve between Sap Flow Velocity (SFV) and Vapour Pressure Deficit (VPD) in a macadamia tree, using data collected between 9 am and 2 pm. The equation derived from this correlation curve is critical for modelling the predicted Sap Flow Velocity (SFV). Different crops may necessitate distinct calibration periods, and adjustments should be made accordingly.

[0156] **Figure 11** demonstrates an observation that the predicted Sap Flow Velocity (SFV) for morning periods closely aligns with the actual Sap Flow Velocity (SFV) when the tree is not experiencing transpiration stress.

[0157] **Figure 12** illustrates the average predicted of Sap Flow Velocity (SFV) for the morning period (9 am to 2 pm) over a 12 day window from 11th June to 22nd June. No transpiration stress was observed until 17th June. From 18th June to 22nd June, drought-induced transpiration stress is evidenced by the discrepancy between the predicted and actual Sap Flow Velocity (SFV).

[0158] Figures 13 to 16 show examples of application of system 200 to a macadamia tree. The figures illustrate data obtained from a second sensor. Use of weather forecast module 230 from figure 2 is not displayed. It will be appreciated that the examples may be displayed differently when data from weather forecast module 230 is obtained

[0159] **Figure 13** illustrates the relationship between Sap Flow Velocity (SFV) and Vapour Pressure Deficit (VPD) during a four-day calibration period from 6th June 2024 to 9th June 2024, when the macadamia tree was not subjected to transpiration stress. The data indicates a strong positive correlation between SFV and VPD under these conditions, suggesting that Sap Flow Velocity (SFV) increases proportionally with Vapour Pressure Deficit (VPD) in the absence of transpiration stress.

[0160] **Figure 14** depicts the correlation curve between Sap Flow Velocity (SFV) and Vapour Pressure Deficit (VPD) in a macadamia plant, using data collected between 9 am and 2 pm. The equation derived from this correlation curve is critical for modelling the predicted Sap Flow Velocity (SFV). It is important to note that different crops may necessitate distinct calibration periods, and adjustments should be made accordingly.

[0161] **Figure 15** shows that the predicted Sap Flow Velocity (SFV) for morning periods closely aligns with the actual Sap Flow Velocity when the tree is not experiencing transpiration stress.

[0162] **Figure 16** illustrates the prediction of Sap Flow Velocity (SFV) for the morning period (9 am to 2 pm) over a twelve-day window from 11th June to 22nd June. No transpiration stress was observed until 17th June. From 18th June to 22nd June, drought-induced transpiration stress is evidenced by the discrepancy between the predicted and actual Sap Flow Velocity (SFV).

[0163] **Figure 17** illustrates the influence that other environmental variables may have on Sap Flow Velocity (SFV). One example shown in figure 17 is soil temperature.

[0164] **Figure 18** shows an example of a recommended irrigation schedule generated, for example, by report generator 240 of irrigation management system 200. The schedule may be displayed, for example, on output device 1930 of computing device 1900.

[0165] In one example the schedule 1800 comprises an aerial view of at least part of a ground surface 1805. Schedule 1800 may be divided into one or more regions. The regions have the effect of visually grouping together groups of plants that have the same or similar transpiration stress.

[0166] For example, region 1810 indicates a group of plants categorised as having waterlogged high stress. Region 1815 indicates a group of plants categorised as having waterlogged low stress. Region 1820 indicates a group of plants categorised as having no transpiration stress. Region 1825 indicates a group of plants categorised as having drought low stress. An example of a presentation style for a region having drought high stress is indicated at 1830.

[0167] The regions may be presented in a visually distinctive manner from each other to enable a user viewing the recommended irrigation schedule to implement different irrigation schedules specifically for the respective regions.

[0168] **Figure 19** illustrates an example of a computing device 1900 configured to implement components of one or more embodiments described above. Examples of such components

include predicted sap flow module 210, TSI module 220, weather forecast module 230, report generator 240, and/or controller module 250.

[0169] In one configuration, computing device 1900 includes at least one processing unit 1905 and computer readable media such as memory 1910. Depending on the exact configuration and type of computing device, memory 1910 may be volatile (such as RAM, for example), non-volatile (such as ROM, flash memory, etc., for example) or some combination of the two.

[0170] In other embodiments, computing device 1900 includes additional features and/or functionality. For example, device 1900 may also include additional storage (e.g., removable and/or non-removable) including, but not limited to, magnetic storage, optical storage, and the like. Such additional storage is illustrated in figure 19 by computer readable media such as storage 1915.

[0171] In an embodiment, computer readable instructions to implement components of one or more embodiments provided herein reside in storage 1915. In an embodiment, storage 1915 also stores other computer readable instructions to implement an operating system, an application program, and the like. Computer readable instructions may be loaded in memory 1910 for execution by processing unit 1905, for example.

[0172] Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, Digital Versatile Disks (DVDs) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium configured to store the desired information and configured to be accessed by computing device 1900. Any such computer storage media may be part of device 1900.

[0173] Computing device 1900 may include communication connection(s), shown as Comms 1920, that allows device 1900 to communicate with other devices. Communication connection(s) 1920 may include, but is not limited to, a modem, a Network Interface Card (NIC), an integrated network interface, a radio frequency transmitter/receiver, an infrared port, a USB connection, or other interfaces for connecting computing device 1900 to other computing devices. Communication connection(s) 1920 may include a wired connection or a wireless connection. Communication connection(s) 1920 may transmit and/or receive communication media.

[0174] Device 1900 may include input device(s) 1925 such as keyboard, mouse, pen, voice input device, touch input device, infrared cameras, video input devices, and/or any other input device. Output device(s) 1930 such as one or more displays, speakers, printers, and/or any other output device may also be included in device 1900. Input device(s) 1925 and output device(s) 1930 may be connected to device 1900 via a wired connection, wireless connection, or any combination thereof. In one embodiment, an input device or an output device from another computing device may be used as input device(s) 1925 or output device(s) 1930 for computing device 1900.

[0175] In an embodiment, components of computing device 1900 are connected by various interconnects, such as a bus 1935. Such interconnects may include a Peripheral Component Interconnect (PCI), such as PCI Express, a Universal Serial Bus (USB), firewire (IEEE 13104), an optical bus structure, and the like. In another embodiment, components of computing device 1900 are interconnected by a network. For example, memory 1910 may be comprised of multiple physical memory units located in different physical locations interconnected by a network.

[0176] Storage devices used to store computer readable instructions may be distributed across a network. For example, a remote computing device (not shown) similar to device 1900, accessible via a network (not shown), may store computer readable instructions to implement one or more embodiments provided herein. Computing device 1900 may access the remote computing device and download a part or all of the computer readable instructions for execution. Alternatively, computing device 1900 may download pieces of the computer readable instructions, as needed, or some instructions may be executed at computing device 1900 and some at remote computing device.

[0177] The term 'connected to' in relation to data or signal transfer as used in this specification includes all direct or indirect types of communication, including wired and wireless, via a cellular network, via a data bus, or any other computer structure. It is envisaged that they may be intervening elements between the connected integers. Variants such as 'in communication with', 'joined to', and 'attached to' are to be interpreted in a similar manner. Related terms such as 'connecting' and 'in connection with' are to be interpreted in the same manner.

[0178] The terms 'component', 'module', 'system', 'interface', and/or the like as used in this specification in relation to a processor are generally intended to refer to a computer-related

entity, either hardware, a combination of hardware and software, software, or software in execution. For example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a controller and the controller can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers.

[0179] As used herein, the term 'computer-readable medium' should be taken to include a single medium or multiple media. Examples of multiple media include a centralised or distributed database and/or associated caches. These multiple media store the one or more sets of computer executable instructions. The term 'computer readable medium' should also be taken to include any medium that is capable of storing, encoding or carrying a set of instructions for execution by a processor and that cause the processor to perform any one or more of the methods described above. The computer-readable medium is also capable of storing, encoding or carrying data structures used by or associated with these sets of instructions. The term 'computer-readable medium' includes solid-state memories, optical media and magnetic media.

CLAUSES DESCRIBING EMBODIMENTS

1. A method of detecting transpiration stress for at least one plant, the method comprising:
receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;
receiving a plurality of predicted SFV data values associated to the at least one plant;
and
calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.
2. A method of detecting transpiration stress for at least one plant, the method comprising:
receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;
receiving a plurality of environmental data values associated to at least one environmental variable;
determining a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to

the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values; and

calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

3. The method of clause 1 or clause 2 comprising calculating the TSI as a ratio of the predicted SFV data to the measured SFV data.
4. The method of any one of clauses 1 to 3 further comprising:
 - receiving at least one Soil Water Potential (SWP) data value; and
 - assigning either a negative value or a positive value to the calculated TSI based at least partly on a comparison of the at least one SWP with a threshold.
5. The method of clause 4 comprising assigning a negative value to the calculated TSI on determining that the at least one SWP is above the threshold.
6. The method of clause 4 or clause 5 comprising assigning a positive value to the calculated TSI on determining that the at least one SWP is below the threshold.
7. The method of any one of clauses 4 to 6 wherein the threshold is selected from within the range 0 to -30kPa.
8. The method of any one of clauses 1 to 7 wherein:
 - at least some of the predicted SFV data values are associated to respective timestamps;
 - at least some of the measured Sap Flow Velocity (SFV) data values are associated to respective timestamps; and
 - calculating the Transpiration Stress Index (TSI) comprises calculating the TSI at least partly from those measured SFV data values and those predicted SFV data values having timestamps within a bounded time interval.
9. The method of clause 8 wherein the bounded time interval comprises a time interval between 9am and 12pm.
10. The method of clause 8 wherein the bounded time interval comprises a time interval between 9am and 2pm.

11. The method of any one of clauses 1 to 10 further comprising:
generating an irrigation schedule recommendation at least partly from the calculated TSI.
12. The method of any one of clauses 1 to 11 further comprising automatically controlling an irrigation system at least partly based on the calculated TSI.
13. The method of clause 12 wherein automatically controlling the irrigation system comprises controlling the irrigation system to increase irrigation when the TSI exceeds an upper TSI threshold and to cease irrigation when the TSI falls below a lower TSI threshold.
14. The method of any one of clauses 1 to 13 wherein the at least one plant has a stem that is at least 8mm in diameter.
15. A method of calculating predicted Sap Flow Velocity (SFV) for at least one plant, the method comprising:
receiving a plurality of environmental data values associated to at least one environmental variable; and
determining a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.
16. The method of clause 15 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD), the correlation function configured to take as input VPD data values.
17. The method of clause 16 wherein receiving the plurality of environmental data values comprises:
receiving a plurality of VPD data values associated to VPD.
18. The method of clause 16 wherein receiving the plurality of environmental data values comprises:
receiving a plurality of humidity data values;
receiving a plurality of temperature data values; and

deriving a plurality of VPD data values at least partly from the humidity data values and the temperature data values.

19. The method of clause 15 wherein the at least one environmental variable comprises Evapotranspiration (ET), the correlation function configured to take as input ET data values.

20. The method of clause 19 wherein receiving the plurality of environmental data values comprises:

receiving a plurality of ET data values associated to ET.

21. The method of clause 19 wherein receiving the plurality of environmental data values comprises:

receiving a plurality of humidity data values;

receiving a plurality of temperature data values;

receiving a plurality of solar radiation data values;

receiving a plurality of wind speed data values; and

deriving a plurality of ET data values at least partly from the humidity data values, the temperature data values, the solar radiation data values, and the wind speed data values.

22. The method of any one of clauses 16 to 18 wherein the at least one environmental variable further comprises soil temperature, the correlation function configured to take as input VPD data values and soil temperature data values.

23. The method of any one of clauses 16 to 18 wherein the at least one environmental variable further comprises solar radiation, the correlation function configured to take as input VPD data values and solar radiation data values.

24. The method of any one of clauses 15 to 23 wherein:

at least some of the environmental data values are associated to respective timestamps; and determining the predicted SFV comprises the correlation function taking as input those environmental data values having timestamps within a bounded time interval.

25. The method of clause 24 wherein the bounded time interval comprises a time interval between 9am and 12pm.

26. The method of clause 24 wherein the bounded time interval comprises a time interval

between 9am and 2pm.

27. The method of any one of clauses 15 to 26 wherein the at least one plant has a stem that is at least 8mm in diameter.

28. A method of determining a correlation function for calculating predicted Sap Flow Velocity (SFV) for at least one plant, the method comprising:

receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;

receiving a plurality of environmental data values associated to at least one environmental variable; and

determining a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

29. The method of clause 28 wherein:

at least some of the measured Sap Flow Velocity (SFV) data values are associated to respective timestamps;

at least some of the environmental data values are associated to respective timestamps; and

determining the correlation function comprises correlating those measured SFV data values and environmental data values having timestamps within a bounded time interval.

30. The method of clause 29 wherein the bounded time interval comprises a time interval between 9am and 12pm.

31. The method of clause 29 wherein the bounded time interval comprises a time interval between 9am and 2pm.

32. The method of any one of clauses 28 to 31 wherein determining the correlation function comprises correlating measured SFV data values and environmental data values measured in the absence of transpiration stress of the at least one plant.

33. The method of any one of clauses 28 to 32 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD), the environmental data values comprising a plurality of VPD data values.

34. The method of any one of clauses 28 to 32 wherein the at least one environmental variable comprises Evapotranspiration (ET), the environmental data values comprising a plurality of ET data values.
35. The method of any one of clauses 28 to 32 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD) and Evapotranspiration (ET), the environmental data values comprising a plurality of VPD data values and ET data values.
36. The method of any one of clauses 28 to 35 wherein the at least one plant has a stem that is at least 8mm in diameter.
37. A transpiration stress detection system for at least one plant, the system comprising:
a Transpiration Stress Index (TSI) module configured to:
 receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant,
 receive a plurality of predicted SFV data values associated to the at least one plant, and
 calculate a TSI at least partly from the measured SFV data values and the predicted SFV data values; and
an output device configured to output the calculated TSI.
38. A transpiration stress detection system for at least one plant, the system comprising:
a predicted sap flow module configured to:
 receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant,
 receive a plurality of environmental data values associated to at least one environmental variable, and
 determine a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate the measured SFV data values associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values;
a Transpiration Stress Index (TSI) module configured to:
 receive the plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant,

receive the plurality of predicted SFV data values determined by the predicted sap flow module, and

calculate a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values; and

an output device configured to output the calculated TSI.

39. The system of clause 37 or clause 38 wherein the TSI module is further configured to calculate the TSI as a ratio of the predicted SFV data to the measured SFV data.

40. The system of any one of clauses 37 or clause 39 wherein the TSI module is further configured to:

receive at least one Soil Water Potential (SWP) data value; and

assign either a negative value or a positive value to the calculated TSI based at least partly on a comparison of the at least one SWP with a threshold.

41. The system of clause 40 wherein the TSI module is configured to assign a negative value to the calculated TSI on determining that the at least one SWP is above the threshold.

42. The system of clause 40 or clause 41 wherein the TSI module is configured to assign a positive value to the calculated TSI on determining that the at least one SWP is below the threshold.

43. The system of any one of clauses 40 to 42 wherein the threshold is selected from within the range 0 to -30kPa.

44. The system of any one of clauses 37 to 43 wherein:

at least some of the predicted SFV data values are associated to respective timestamps;

at least some of the measured Sap Flow Velocity (SFV) data values are associated to respective timestamps; and

calculating the Transpiration Stress Index (TSI) comprises calculating the TSI at least partly from those measured SFV data values and those predicted SFV data values having timestamps within a bounded time interval.

45. The system of clause 44 wherein the bounded time interval comprises a time interval between 9am and 12pm.

46. The system of clause 44 wherein the bounded time interval comprises a time interval between 9am and 2pm.
47. The system of any one of clauses 37 to 46 further comprising:
a report generator configured to generate an irrigation schedule recommendation at least partly from the calculated TSI.
48. The system of any one of clauses 37 to 47 further comprising:
a controller module configured to automatically control an irrigation system at least partly based on the calculated TSI.
49. The system of clause 48 wherein automatically controlling the irrigation system comprises controlling the irrigation system to increase irrigation when the TSI exceeds an upper TSI threshold and to cease irrigation when the TSI falls below a lower TSI threshold.
50. The system of any one of clauses 37 to 49 wherein the at least one plant has a stem that is at least 8mm in diameter.
51. A predicted Sap Flow Velocity (SFV) calculation system for at least one plant, the system comprising:
a predicted sap flow module configured to:
receive a plurality of environmental data values associated to at least one environmental variable; and
determine a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.
52. The system of clause 51 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD), the correlation function configured to take as input VPD data values.
53. The system of clause 52 wherein receiving the plurality of environmental data values comprises:
receiving a plurality of VPD data values associated to VPD.

54. The system of clause 52 wherein receiving the plurality of environmental data values comprises:
- receiving a plurality of humidity data values;
 - receiving a plurality of temperature data values; and
 - deriving a plurality of VPD data values at least partly from the humidity data values and the temperature data values.
55. The system of clause 51 wherein the at least one environmental variable comprises Evapotranspiration (ET), the correlation function configured to take as input ET data values.
56. The system of clause 55 wherein receiving the plurality of environmental data values comprises:
- receiving a plurality of ET data values associated to ET.
57. The system of clause 55 wherein receiving the plurality of environmental data values comprises:
- receiving a plurality of humidity data values;
 - receiving a plurality of temperature data values;
 - receiving a plurality of solar radiation data values;
 - receiving a plurality of wind speed data values; and
 - deriving a plurality of ET data values at least partly from the humidity data values, the temperature data values, the solar radiation data values, and the wind speed data values.
58. The system of any one of clauses 52 to 54 wherein the at least one environmental variable further comprises soil temperature, the correlation function configured to take as input VPD data values and soil temperature data values.
59. The system of any one of clauses 52 to 54 wherein the at least one environmental variable further comprises solar radiation, the correlation function configured to take as input VPD data values and solar radiation data values.
60. The system of any one of clauses 51 to 59 wherein:
- at least some of the environmental data values are associated to respective timestamps; and
 - determining the predicted SFV comprises the correlation function taking as input those

environmental data values having timestamps within a bounded time interval.

61. The system of clause 60 wherein the bounded time interval comprises a time interval between 9am and 12pm.

62. The system of clause 60 wherein the bounded time interval comprises a time interval between 9am and 2pm.

63. The system of any one of clauses 51 to 62 wherein the at least one plant has a stem that is at least 8mm in diameter.

64. A correlation function determination system for calculating predicted Sap Flow Velocity (SFV) for at least one plant, the system comprising:

a predicted sap flow module configured to:

receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;

receive a plurality of environmental data values associated to at least one environmental variable; and

determine a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

65. The system of clause 64 wherein:

at least some of the measured Sap Flow Velocity (SFV) data values are associated to respective timestamps;

at least some of the environmental data values are associated to respective timestamps; and

determining the correlation function comprises correlating those measured SFV data values and environmental data values having timestamps within a bounded time interval.

66. The system of clause 65 wherein the bounded time interval comprises a time interval between 9am and 12pm.

67. The system of clause 65 wherein the bounded time interval comprises a time interval between 9am and 2pm.

68. The system of any one of clauses 64 to 67 wherein determining the correlation function comprises correlating measured SFV data values and environmental data values measured in the absence of transpiration stress of the at least one plant.

69. The system of any one of clauses 64 to 68 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD), the environmental data values comprising a plurality of VPD data values.

70. The system of any one of clauses 64 to 68 wherein the at least one environmental variable comprises Evapotranspiration (ET), the environmental data values comprising a plurality of ET data values.

71. The system of any one of clauses 64 to 68 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD) and Evapotranspiration (ET), the environmental data values comprising a plurality of VPD data values and ET data values.

72. The system of any one of clauses 64 to 71 wherein the at least one plant has a stem that is at least 8mm in diameter.

73. A transpiration stress detection system for at least one plant, the system comprising:
a storage device; and
a processor programmed to:
 receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;
 receive a plurality of predicted SFV data values associated to the at least one plant; and
 calculate a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

74. A transpiration stress detection system for at least one plant, the system comprising:
a storage device; and
a processor programmed to:
 receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant,
 receive a plurality of environmental data values associated to at least one environmental variable,

determine a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values, and
calculate a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

75. A predicted Sap Flow Velocity (SFV) calculation system for at least one plant, the system comprising:

a storage device; and

a processor programmed to:

receive a plurality of environmental data values associated to at least one environmental variable, and

determine a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

76. A correlation function determination system for calculating predicted Sap Flow Velocity (SFV) for at least one plant, the system comprising:

a storage device; and

a processor programmed to:

receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant,

receive a plurality of environmental data values associated to at least one environmental variable, and

determine a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

77. A computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of detecting transpiration stress for at least one plant, the method comprising:

receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;

receiving a plurality of predicted SFV data values associated to the at least one plant;

and

calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

78. A computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of detecting transpiration stress for at least one plant, the method comprising:

receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;

receiving a plurality of environmental data values associated to at least one environmental variable;

determining a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values; and

calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

79. A computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of calculating predicted Sap Flow Velocity (SFV) for at least one plant, the method comprising:

receiving a plurality of environmental data values associated to at least one environmental variable; and

determining a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

80. A computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of determining a correlation function for calculating predicted Sap Flow Velocity (SFV) for at least one plant, the method comprising:

receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;

receiving a plurality of environmental data values associated to at least one environmental variable; and

determining a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

81. A method of detecting soil drought stress for a type of plant, the method comprising: calculating a drought stress calibration curve for the type of plant comprising: installing sap flow sensors in a plurality of plants of the type, obtaining sap flow velocity data from the sap flow sensors, measuring temperature and humidity data on-site, and calculating the drought stress calibration curve using the sap flow velocity data and the temperature and humidity data; and detecting soil drought stress for the type of plant by: obtaining live sap flow velocity data from the sap flow sensors, calculating a predicted sap flow using live temperature and humidity (VPD) data and the drought stress calibration curve, and detecting a soil drought stress index as the residual between the live sap flow velocity data and the predicted sap flow.

82. The method as claimed in clause 81, wherein a drought stress calibration curve is calculated for each plant and wherein the method further comprises calculating a drought stress calibration curve from each drought stress calibration curve.

83. The method as claimed in clause 81, wherein calculating the drought stress calibration curve is performed during a well-watered period.

84. The method as claimed in clause 81, further comprising automatically controlling an irrigation system according to the soil drought stress index.

85. The method as claimed in clause 84, wherein automatically controlling the irrigation system comprises controlling the irrigation system to increase irrigation when the soil drought stress index falls beneath a threshold and to cease increasing irrigation when the soil drought stress index exceeds the threshold.

CLAIMS

1. A method of detecting transpiration stress for at least one plant, the method comprising:
receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;
receiving a plurality of predicted SFV data values associated to the at least one plant;
and
calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.
2. The method of claim 1 comprising calculating the TSI as a ratio of the predicted SFV data to the measured SFV data.
3. The method of claim 1 or claim 2 further comprising:
receiving at least one Soil Water Potential (SWP) data value; and
assigning either a negative value or a positive value to the calculated TSI based at least partly on a comparison of the at least one SWP with a threshold.
4. The method of any one of claims 1 to 3 wherein:
at least some of the predicted SFV data values are associated to respective timestamps;
at least some of the measured Sap Flow Velocity (SFV) data values are associated to respective timestamps; and
calculating the Transpiration Stress Index (TSI) comprises calculating the TSI at least partly from those measured SFV data values and those predicted SFV data values having timestamps within a bounded time interval.
5. The method of any one of claims 1 to 4 further comprising:
generating an irrigation schedule recommendation at least partly from the calculated TSI.
6. The method of any one of claims 1 to 5 further comprising automatically controlling an irrigation system at least partly based on the calculated TSI.
7. The method of any one of claims 1 to 6 wherein the at least one plant has a stem that is at least 8mm in diameter.

8. A method of calculating predicted Sap Flow Velocity (SFV) for at least one plant, the method comprising:
- receiving a plurality of environmental data values associated to at least one environmental variable; and
 - determining a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.
9. The method of claim 8 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD), the correlation function configured to take as input VPD data values.
10. The method of claim 8 wherein the at least one environmental variable comprises Evapotranspiration (ET), the correlation function configured to take as input ET data values.
11. The method of claim 9 wherein the at least one environmental variable further comprises soil temperature, the correlation function configured to take as input VPD data values and soil temperature data values.
12. The method of claim 9 wherein the at least one environmental variable further comprises solar radiation, the correlation function configured to take as input VPD data values and solar radiation data values.
13. The method of any one of claims 8 to 12 wherein:
- at least some of the environmental data values are associated to respective timestamps; and
 - determining the predicted SFV comprises the correlation function taking as input those environmental data values having timestamps within a bounded time interval.
14. The method of any one of claims 8 to 13 wherein the at least one plant has a stem that is at least 8mm in diameter.
15. A method of determining a correlation function for calculating predicted Sap Flow Velocity (SFV) for at least one plant, the method comprising:

receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;

receiving a plurality of environmental data values associated to at least one environmental variable; and

determining a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

16. The method of claim 15 wherein:

at least some of the measured Sap Flow Velocity (SFV) data values are associated to respective timestamps;

at least some of the environmental data values are associated to respective timestamps; and

determining the correlation function comprises correlating those measured SFV data values and environmental data values having timestamps within a bounded time interval.

17. The method of claim 15 or claim 16 wherein determining the correlation function comprises correlating measured SFV data values and environmental data values measured in the absence of transpiration stress of the at least one plant.

18. The method of any one of claims 15 to 17 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD), the environmental data values comprising a plurality of VPD data values.

19. The method of any one of claims 15 to 17 wherein the at least one environmental variable comprises Evapotranspiration (ET), the environmental data values comprising a plurality of ET data values.

20. The method of any one of claims 15 to 17 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD) and Evapotranspiration (ET), the environmental data values comprising a plurality of VPD data values and ET data values.

21. The method of any one of claims 15 to 20 wherein the at least one plant has a stem that is at least 8mm in diameter.

22. A transpiration stress detection system for at least one plant, the system comprising:

a Transpiration Stress Index (TSI) module configured to:
 receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant,
 receive a plurality of predicted SFV data values associated to the at least one plant, and
 calculate a TSI at least partly from the measured SFV data values and the predicted SFV data values; and
 an output device configured to output the calculated TSI.

23. The system of claim 22 wherein the TSI module is further configured to calculate the TSI as a ratio of the predicted SFV data to the measured SFV data.

24. The system of claim 22 or claim 23 wherein the TSI module is further configured to:
 receive at least one Soil Water Potential (SWP) data value; and
 assign either a negative value or a positive value to the calculated TSI based at least partly on a comparison of the at least one SWP with a threshold.

25. The system of any one of claims 22 to 24 wherein:
 at least some of the predicted SFV data values are associated to respective timestamps;
 at least some of the measured Sap Flow Velocity (SFV) data values are associated to respective timestamps; and
 calculating the Transpiration Stress Index (TSI) comprises calculating the TSI at least partly from those measured SFV data values and those predicted SFV data values having timestamps within a bounded time interval.

26. The system of any one of claims 22 to 25 further comprising:
 a report generator configured to generate an irrigation schedule recommendation at least partly from the calculated TSI.

27. The system of any one of claims 22 to 26 further comprising:
 a controller module configured to automatically control an irrigation system at least partly based on the calculated TSI.

28. The system of any one of claims 22 to 27 wherein the at least one plant has a stem that is at least 8mm in diameter.

29. A predicted Sap Flow Velocity (SFV) calculation system for at least one plant, the system comprising:
- a predicted sap flow module configured to:
 - receive a plurality of environmental data values associated to at least one environmental variable; and
 - determine a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.
30. The system of claim 29 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD), the correlation function configured to take as input VPD data values.
31. The system of claim 29 wherein the at least one environmental variable comprises Evapotranspiration (ET), the correlation function configured to take as input ET data values.
32. The system of claim 30 wherein the at least one environmental variable further comprises soil temperature, the correlation function configured to take as input VPD data values and soil temperature data values.
33. The system of claim 30 wherein the at least one environmental variable further comprises solar radiation, the correlation function configured to take as input VPD data values and solar radiation data values.
34. The system of any one of claims 29 to 33 wherein:
- at least some of the environmental data values are associated to respective timestamps; and
 - determining the predicted SFV comprises the correlation function taking as input those environmental data values having timestamps within a bounded time interval.
35. The system of any one of claims 29 to 34 wherein the at least one plant has a stem that is at least 8mm in diameter.
36. A correlation function determination system for calculating predicted Sap Flow Velocity

(SFV) for at least one plant, the system comprising:

a predicted sap flow module configured to:

receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;

receive a plurality of environmental data values associated to at least one environmental variable; and

determine a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

37. The system of claim 36 wherein:

at least some of the measured Sap Flow Velocity (SFV) data values are associated to respective timestamps;

at least some of the environmental data values are associated to respective timestamps; and

determining the correlation function comprises correlating those measured SFV data values and environmental data values having timestamps within a bounded time interval.

38. The system of claim 36 or claim 37 wherein determining the correlation function comprises correlating measured SFV data values and environmental data values measured in the absence of transpiration stress of the at least one plant.

39. The system of any one of claims 36 to 38 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD), the environmental data values comprising a plurality of VPD data values.

40. The system of any one of claims 36 to 38 wherein the at least one environmental variable comprises Evapotranspiration (ET), the environmental data values comprising a plurality of ET data values.

41. The system of any one of claims 36 to 38 wherein the at least one environmental variable comprises Vapour Pressure Deficit (VPD) and Evapotranspiration (ET), the environmental data values comprising a plurality of VPD data values and ET data values.

42. The system of any one of claims 36 to 41 wherein the at least one plant has a stem that is at least 8mm in diameter.

43. A transpiration stress detection system for at least one plant, the system comprising:
a storage device; and
a processor programmed to:
 receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;
 receive a plurality of predicted SFV data values associated to the at least one plant; and
 calculate a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.
44. A transpiration stress detection system for at least one plant, the system comprising:
a storage device; and
a processor programmed to:
 receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant,
 receive a plurality of environmental data values associated to at least one environmental variable,
 determine a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values, and
 calculate a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.
45. A predicted Sap Flow Velocity (SFV) calculation system for at least one plant, the system comprising:
a storage device; and
a processor programmed to:
 receive a plurality of environmental data values associated to at least one environmental variable, and
 determine a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

46. A correlation function determination system for calculating predicted Sap Flow Velocity (SFV) for at least one plant, the system comprising:

a storage device; and

a processor programmed to:

receive a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant,

receive a plurality of environmental data values associated to at least one environmental variable, and

determine a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

47. A computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of detecting transpiration stress for at least one plant, the method comprising:

receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;

receiving a plurality of predicted SFV data values associated to the at least one plant; and

calculating a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values.

48. A computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of calculating predicted Sap Flow Velocity (SFV) for at least one plant, the method comprising:

receiving a plurality of environmental data values associated to at least one environmental variable; and

determining a predicted SFV for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

49. A computer-readable medium having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to perform a method of determining a correlation function for calculating predicted Sap Flow Velocity (SFV) for at least one plant, the method comprising:

receiving a plurality of measured Sap Flow Velocity (SFV) data values associated to the at least one plant;

receiving a plurality of environmental data values associated to at least one environmental variable; and

determining a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values.

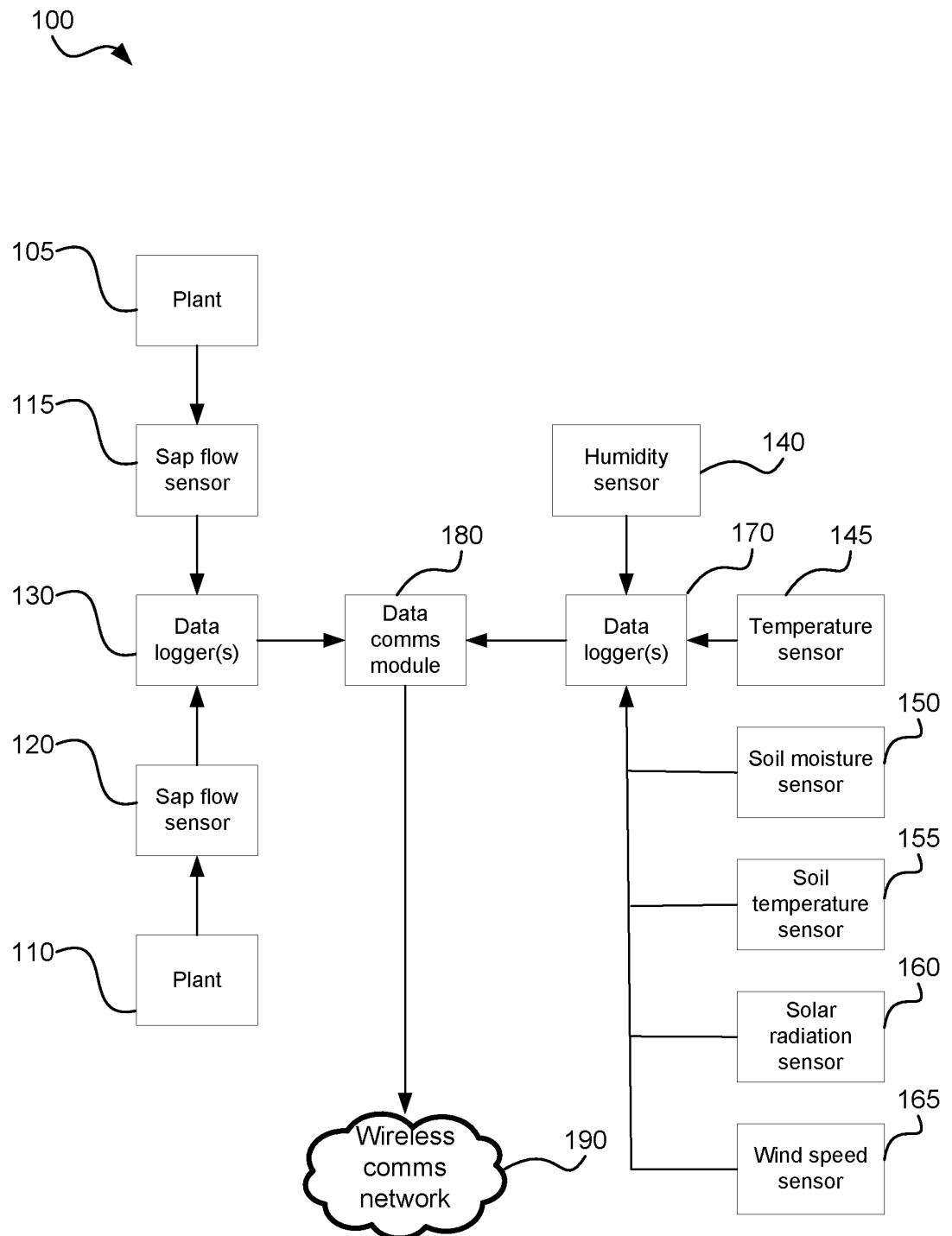


Fig. 1

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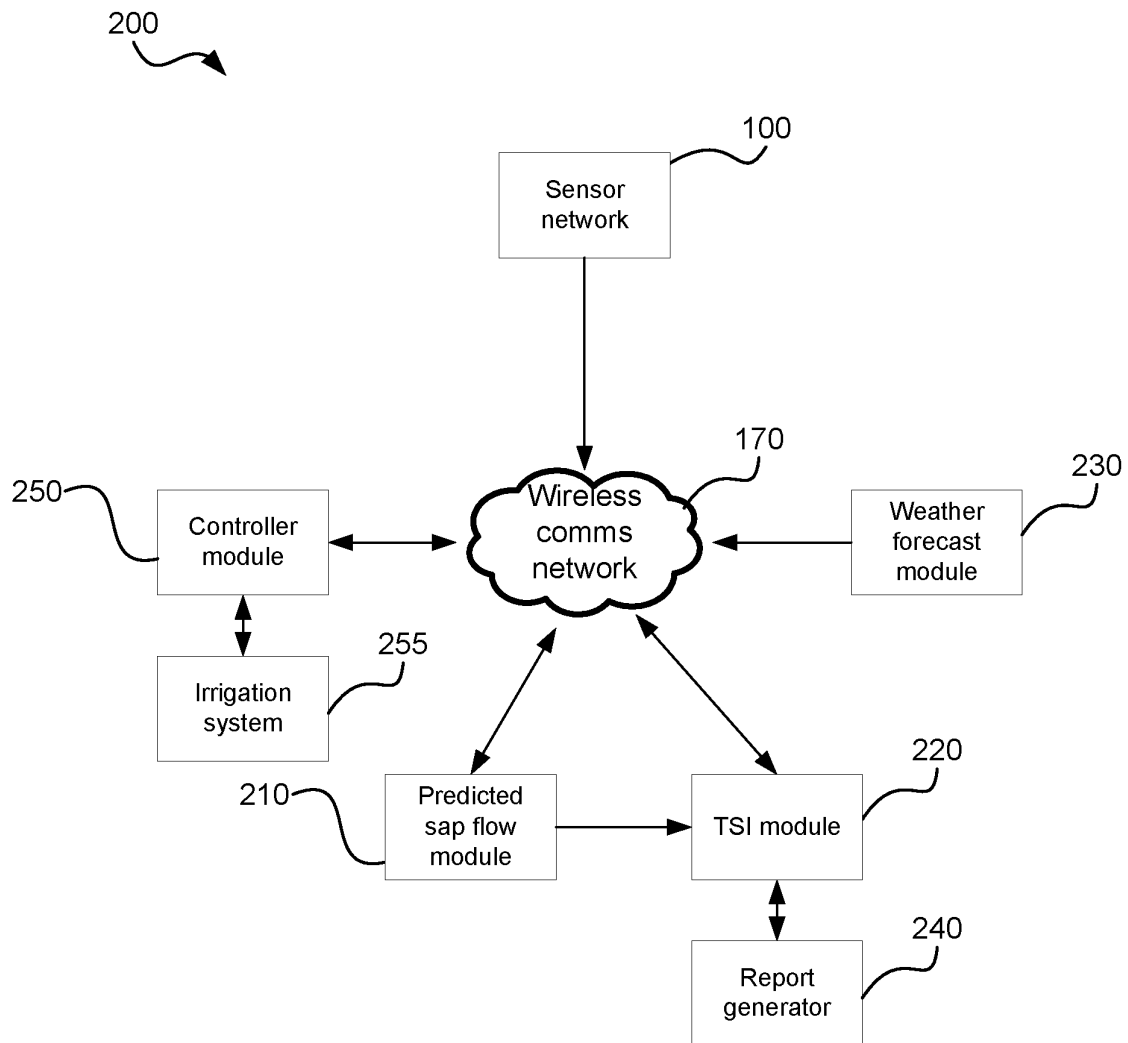


Fig. 2

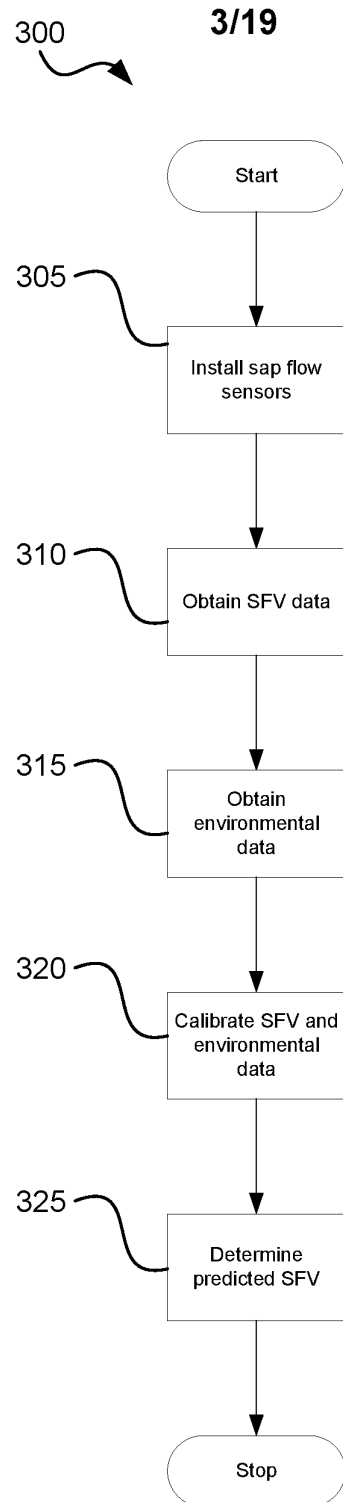


Fig. 3

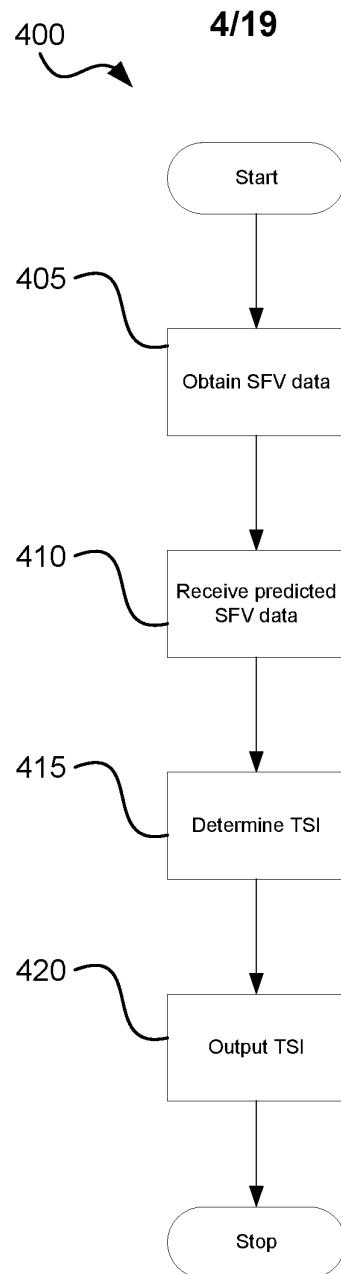


Fig. 4

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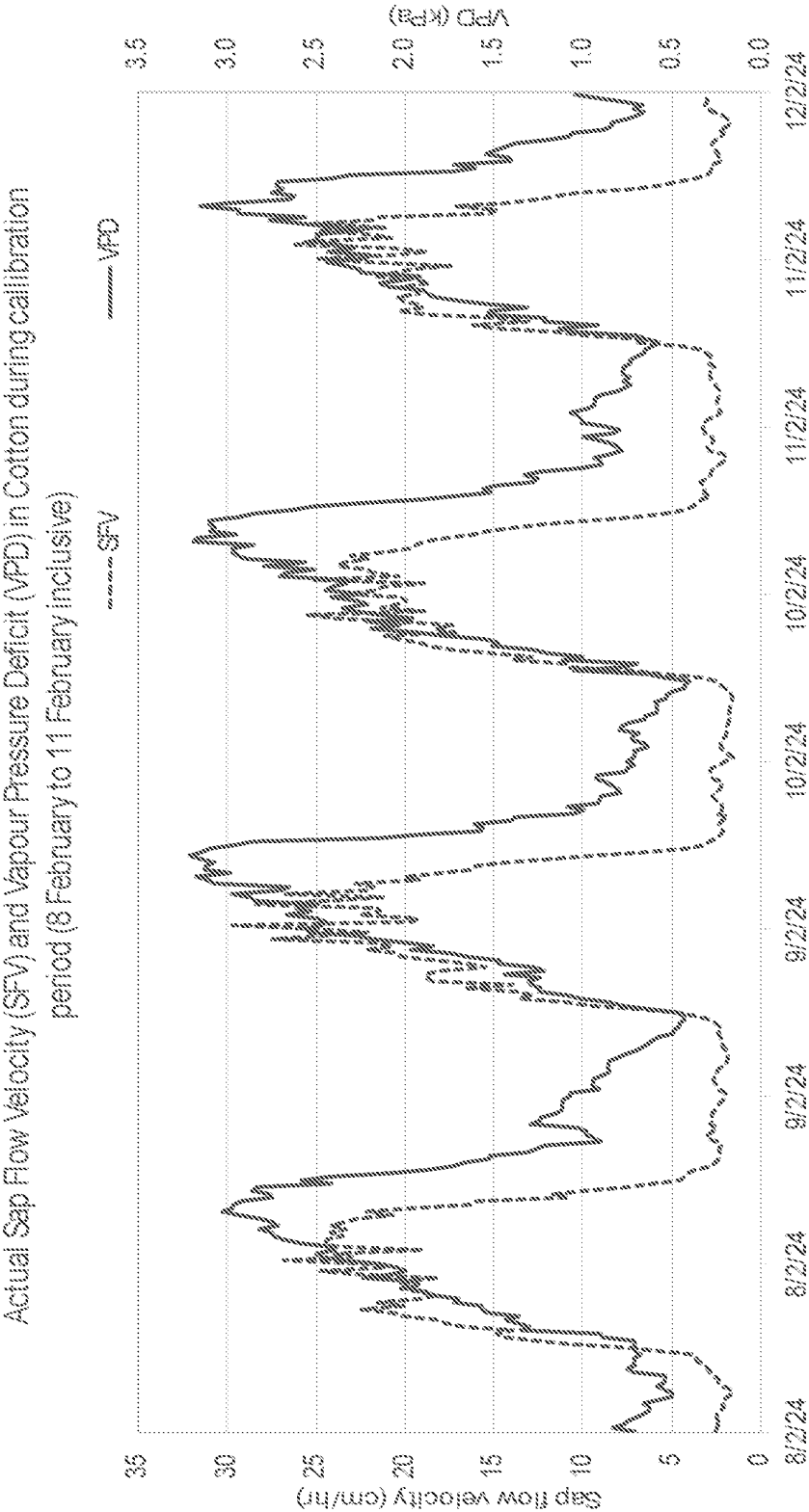


Fig. 5

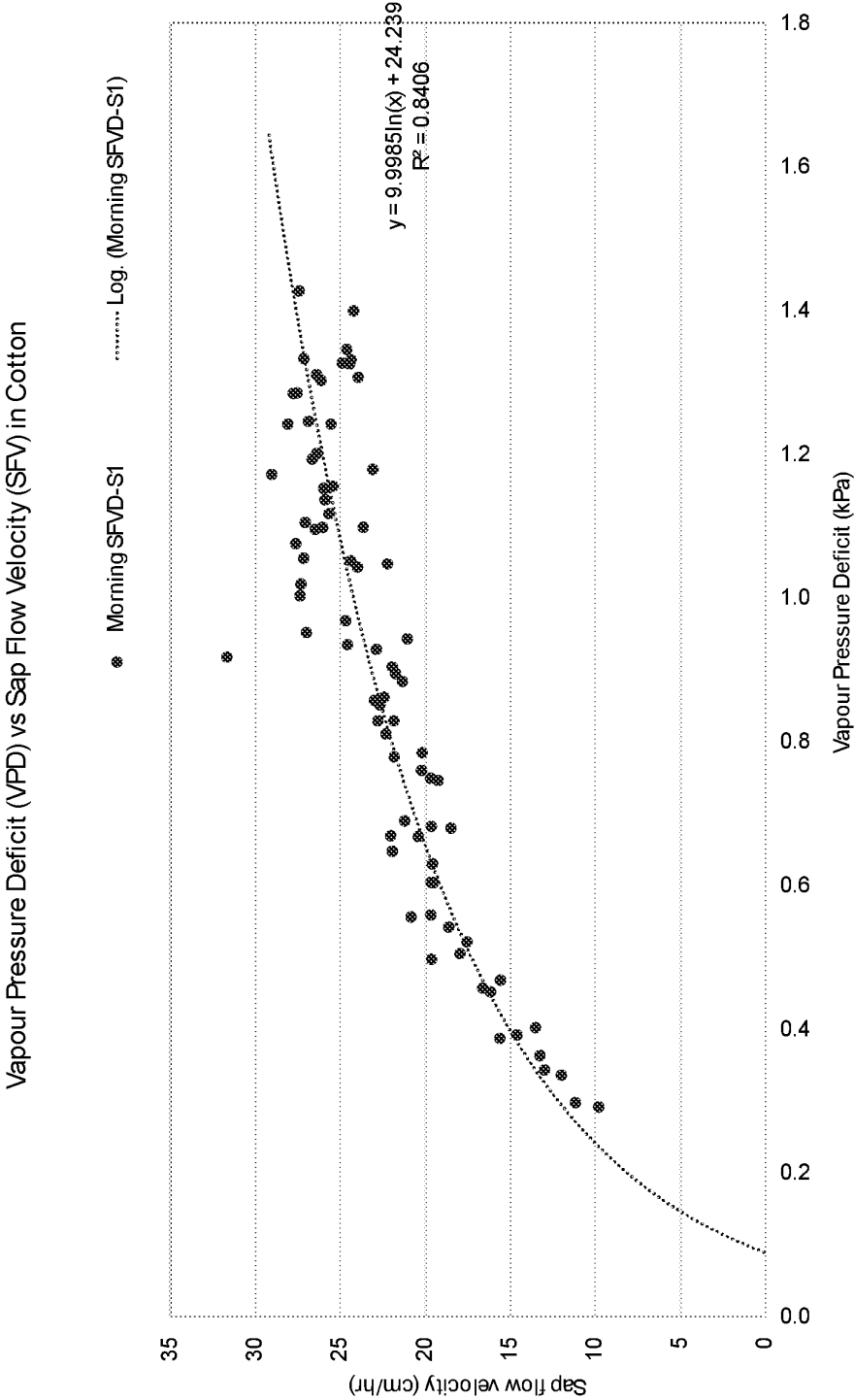


Fig. 6

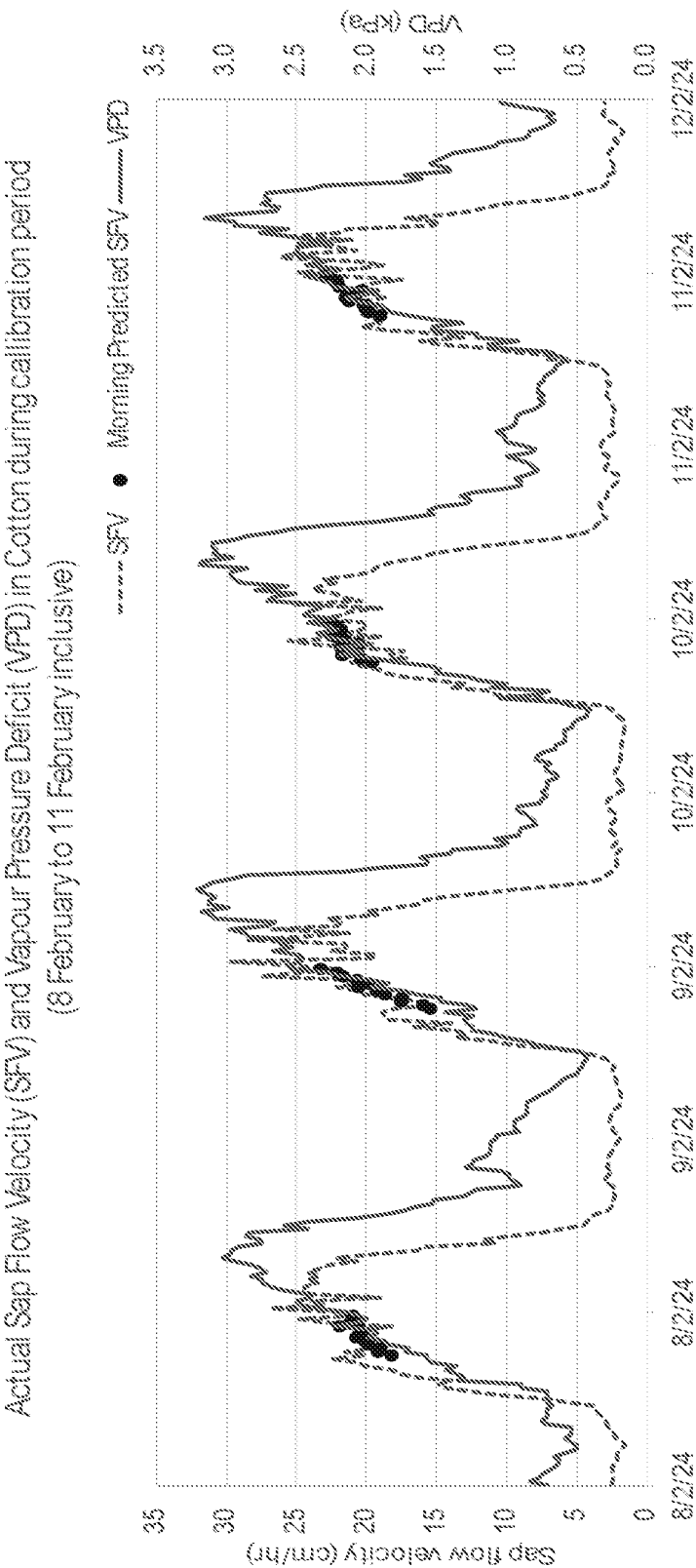


Fig. 7

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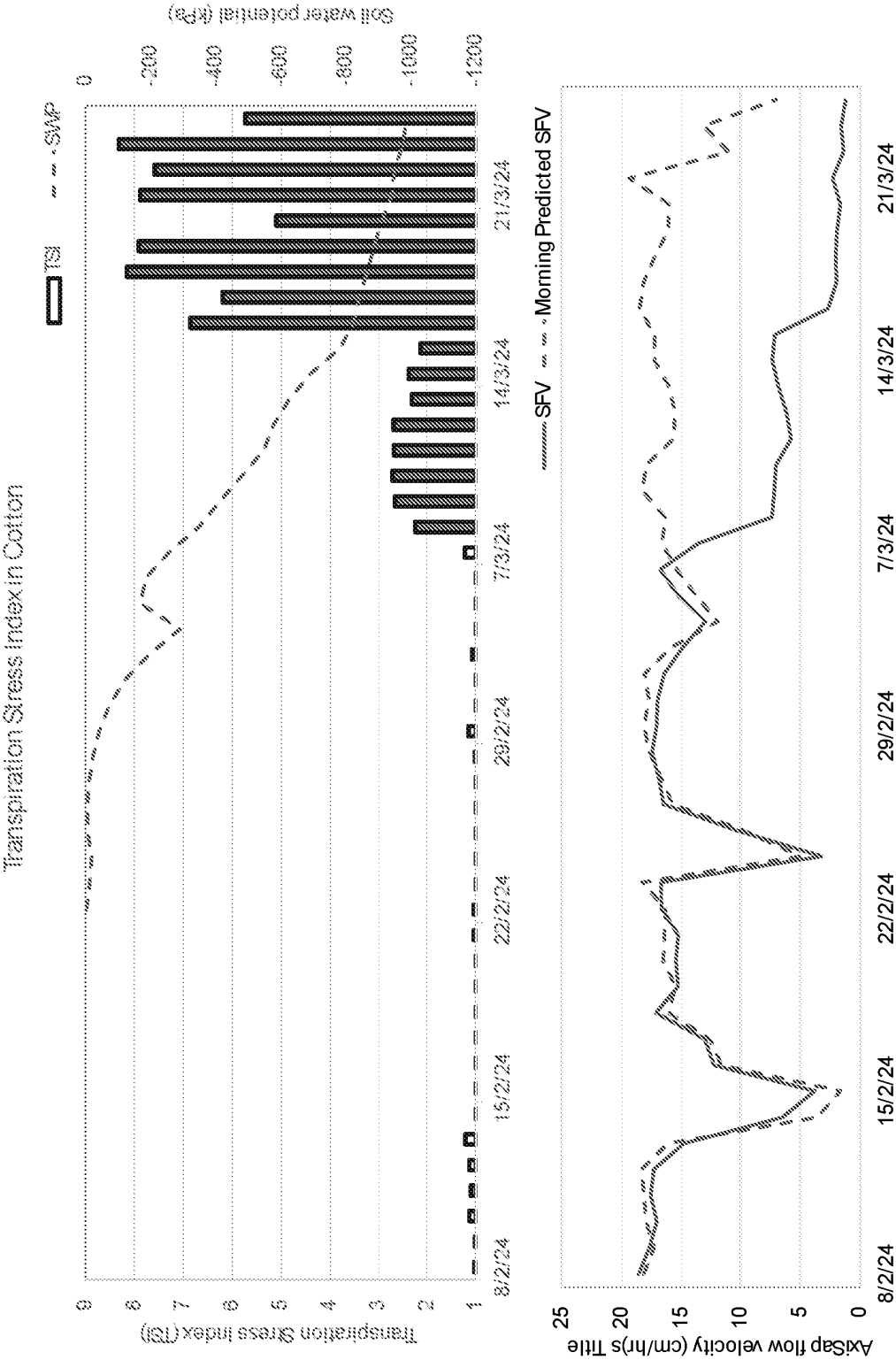


Fig. 8

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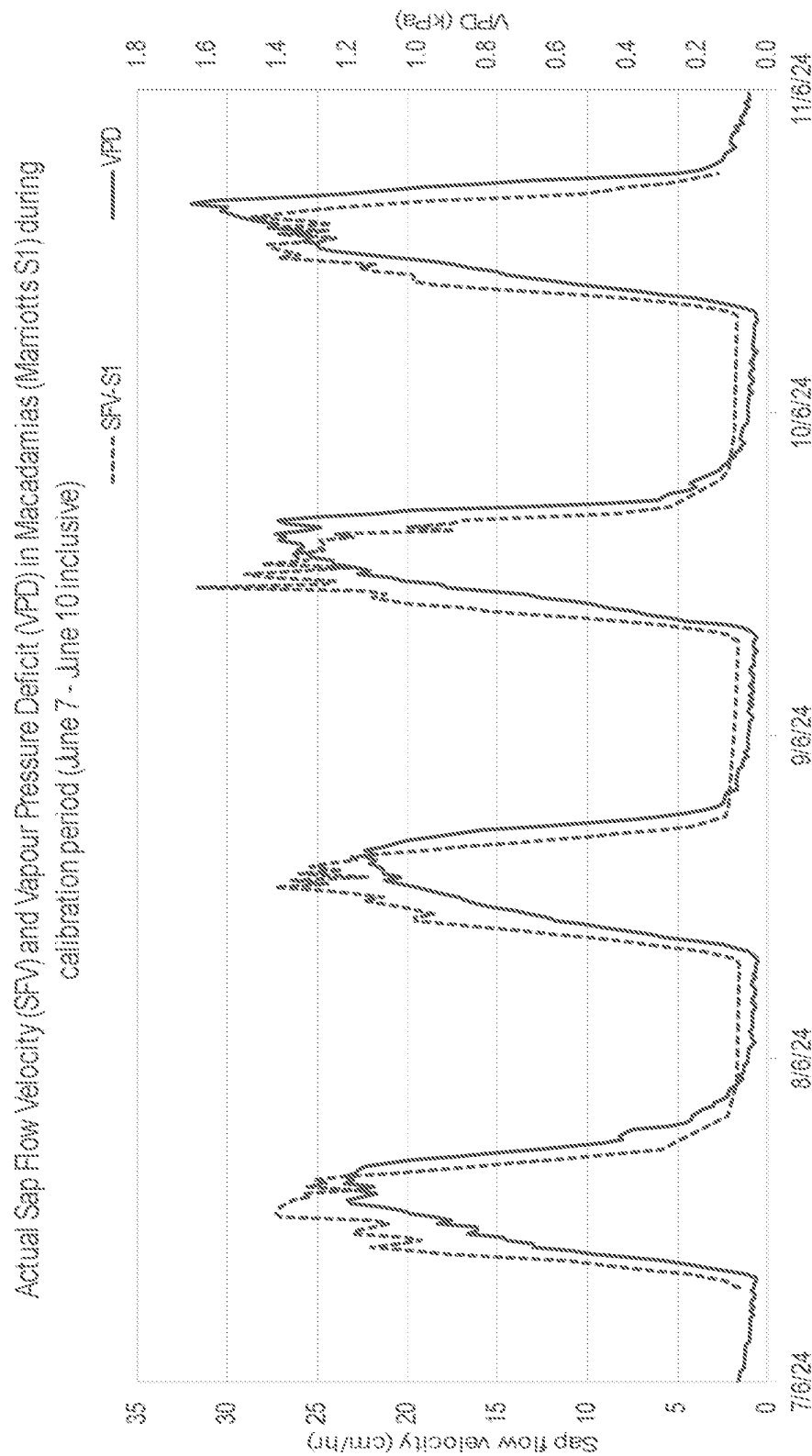


Fig. 9

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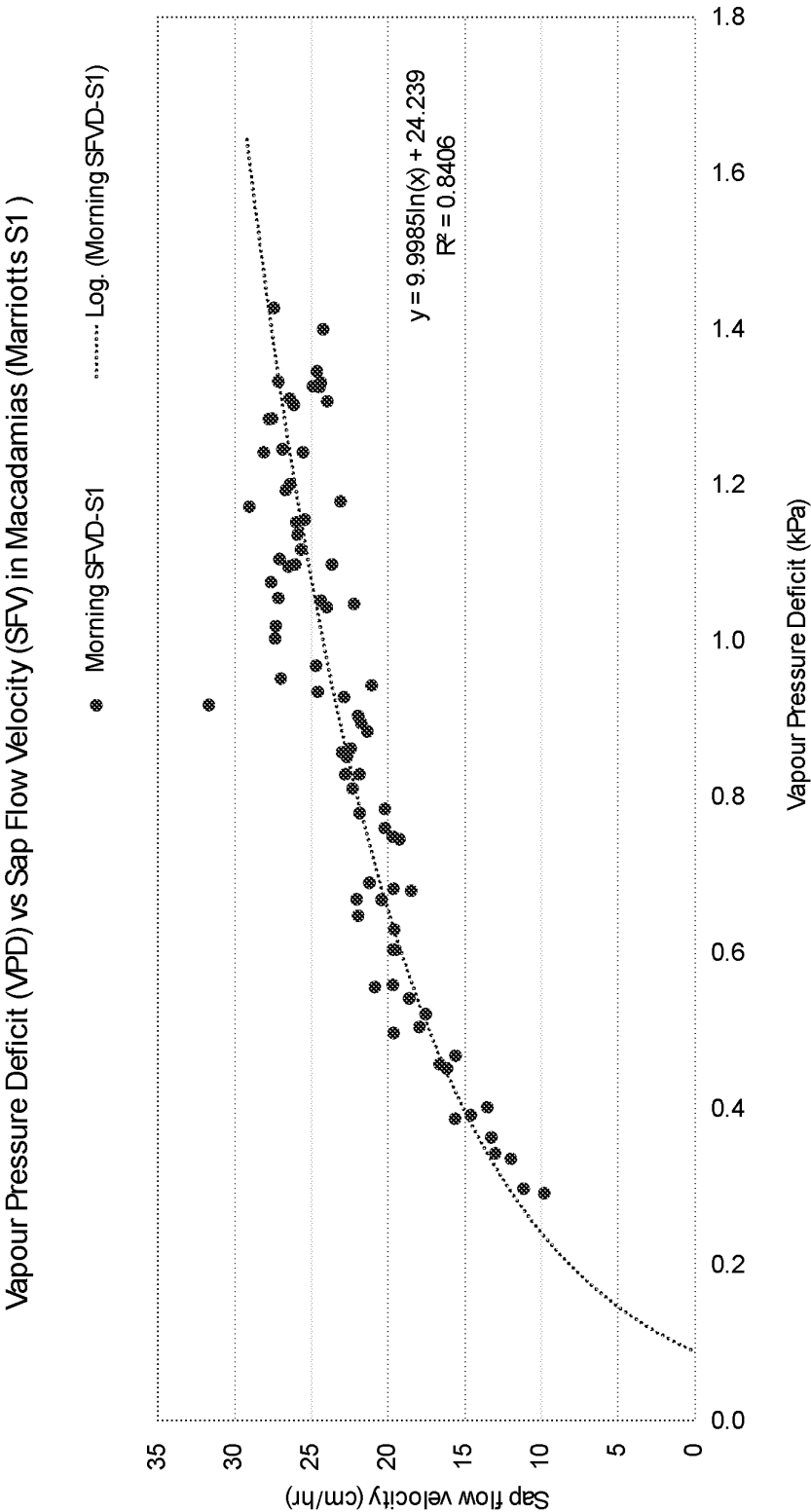


Fig. 10

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Actual Sap Flow Velocity (SFV) and Vapour Pressure Deficit (VPD) in Macadamias (Marrion's S1) during calibration period (June 7 - June 10 inclusive)

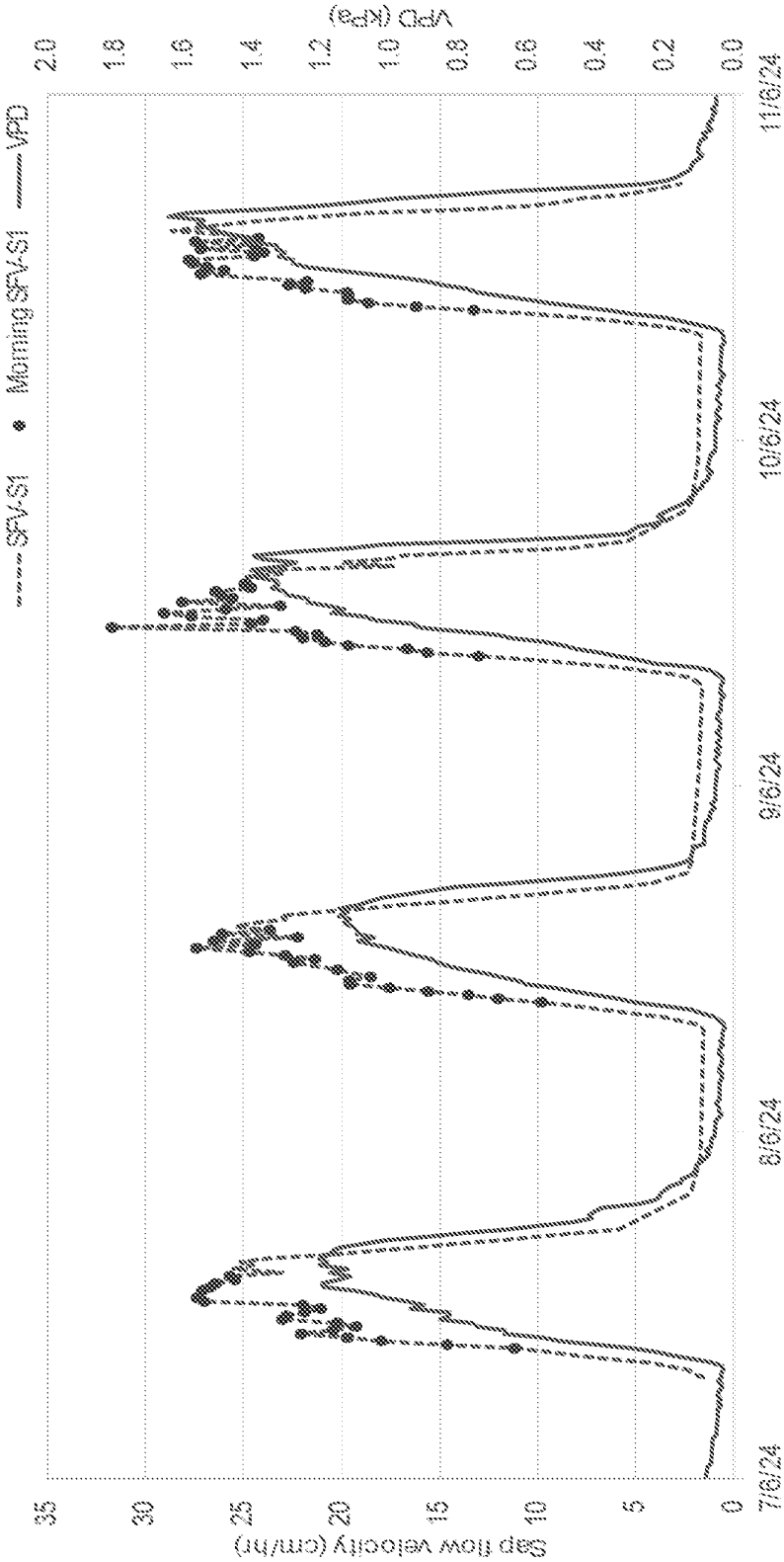


Fig. 11

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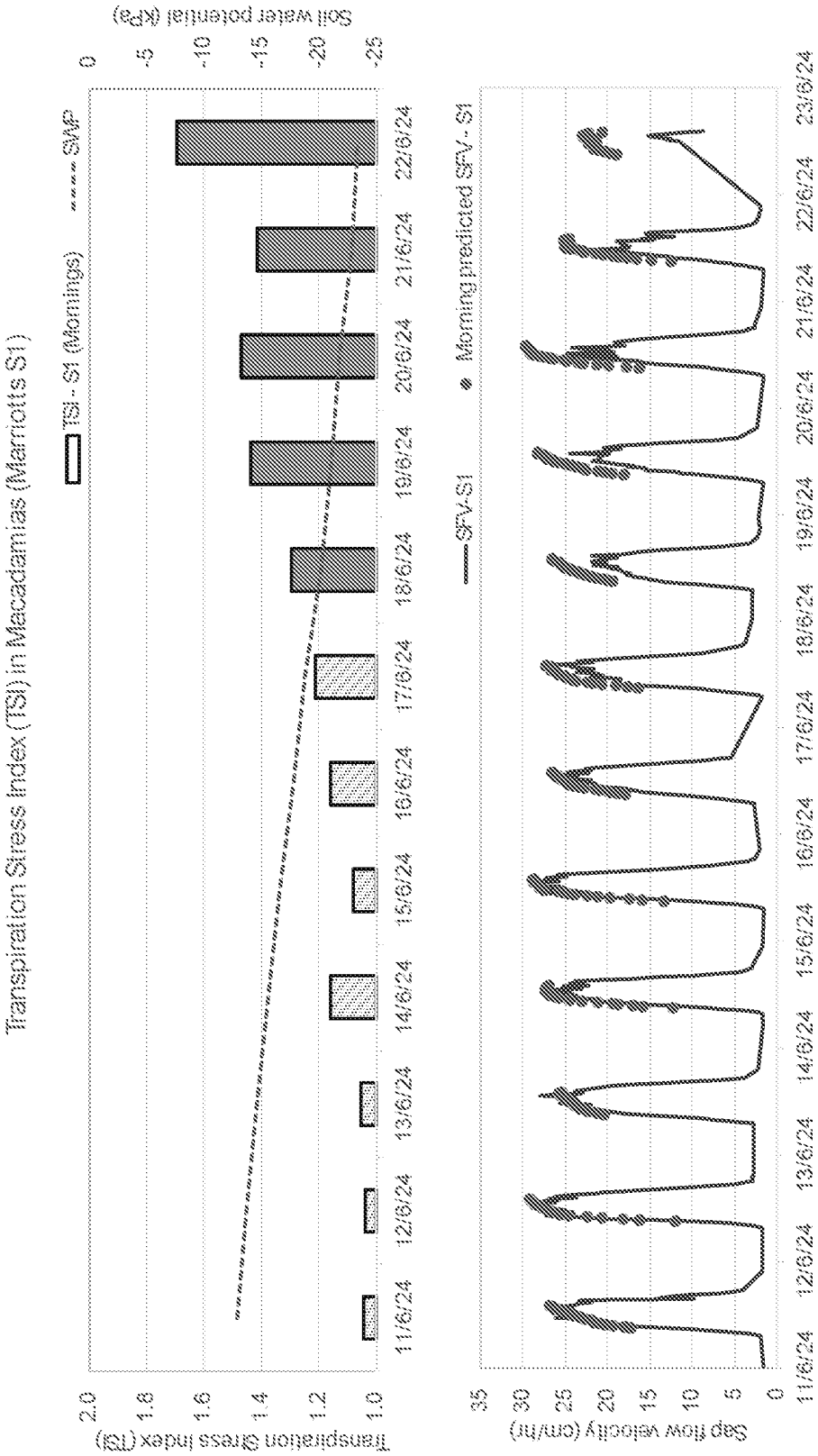


Fig. 12

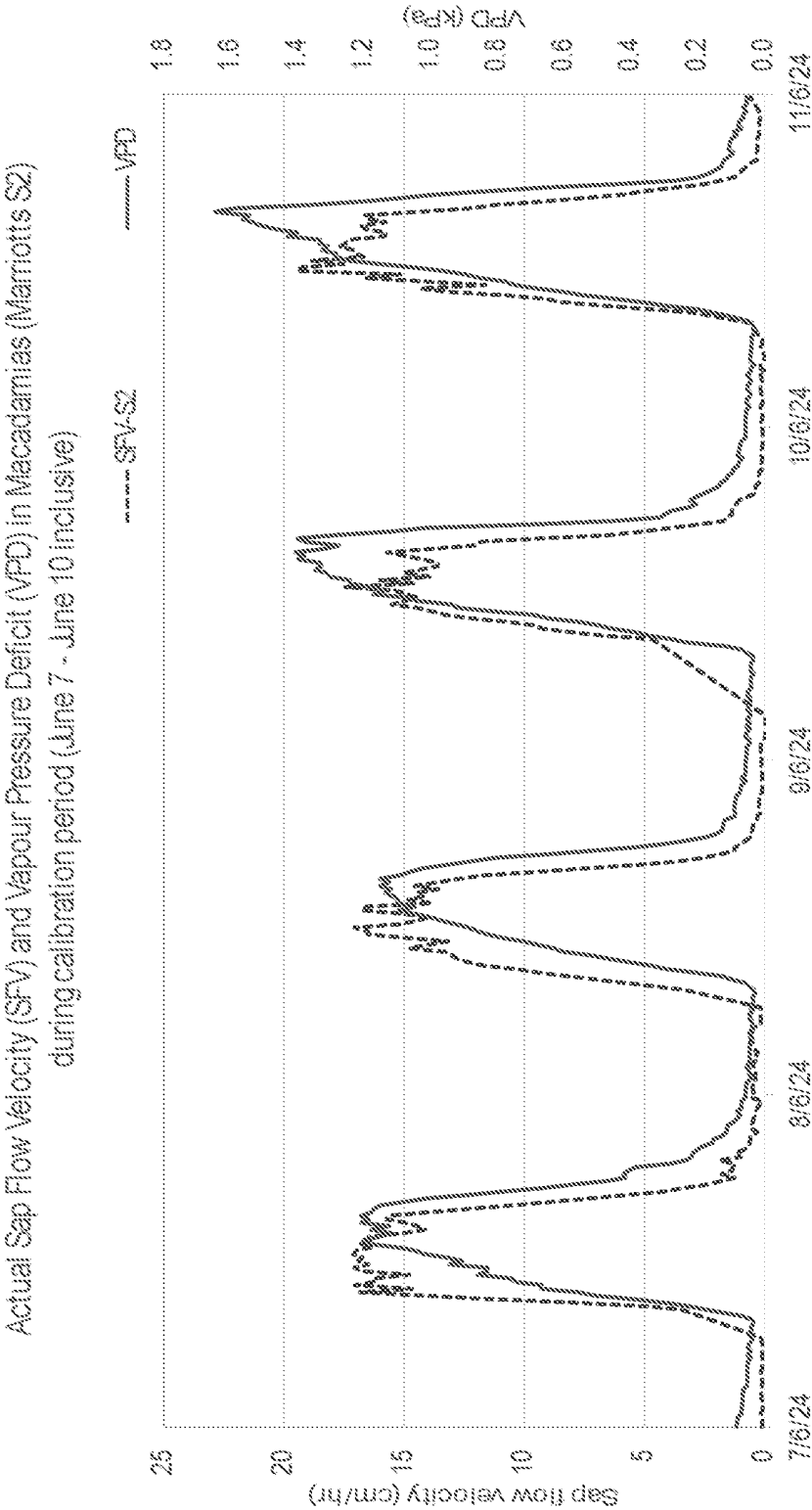


Fig. 13

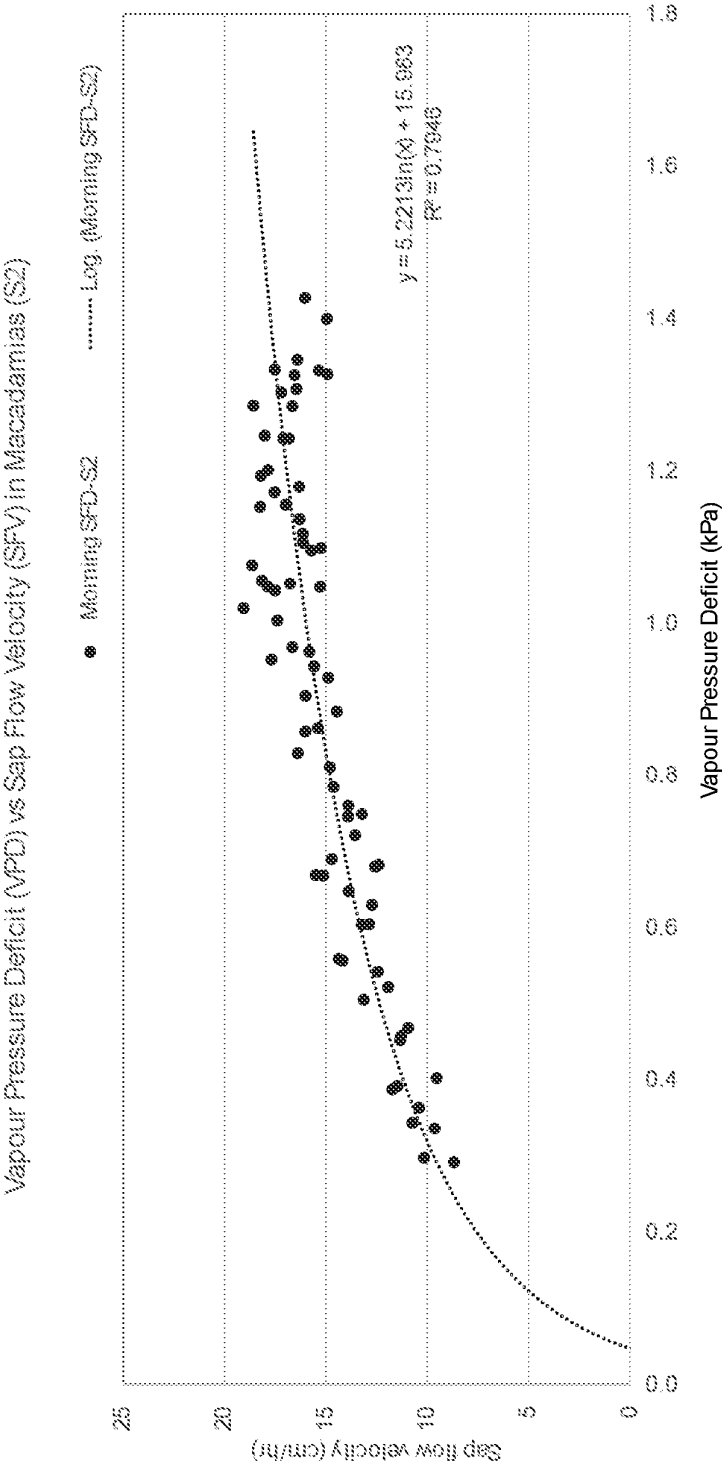


Fig. 14

Actual Sap Flow Velocity (SFV) and Vapour Pressure Deficit (VPD) in Macadamias (Marriotts S2) during calibration period (June 7 - June 10 inclusive)

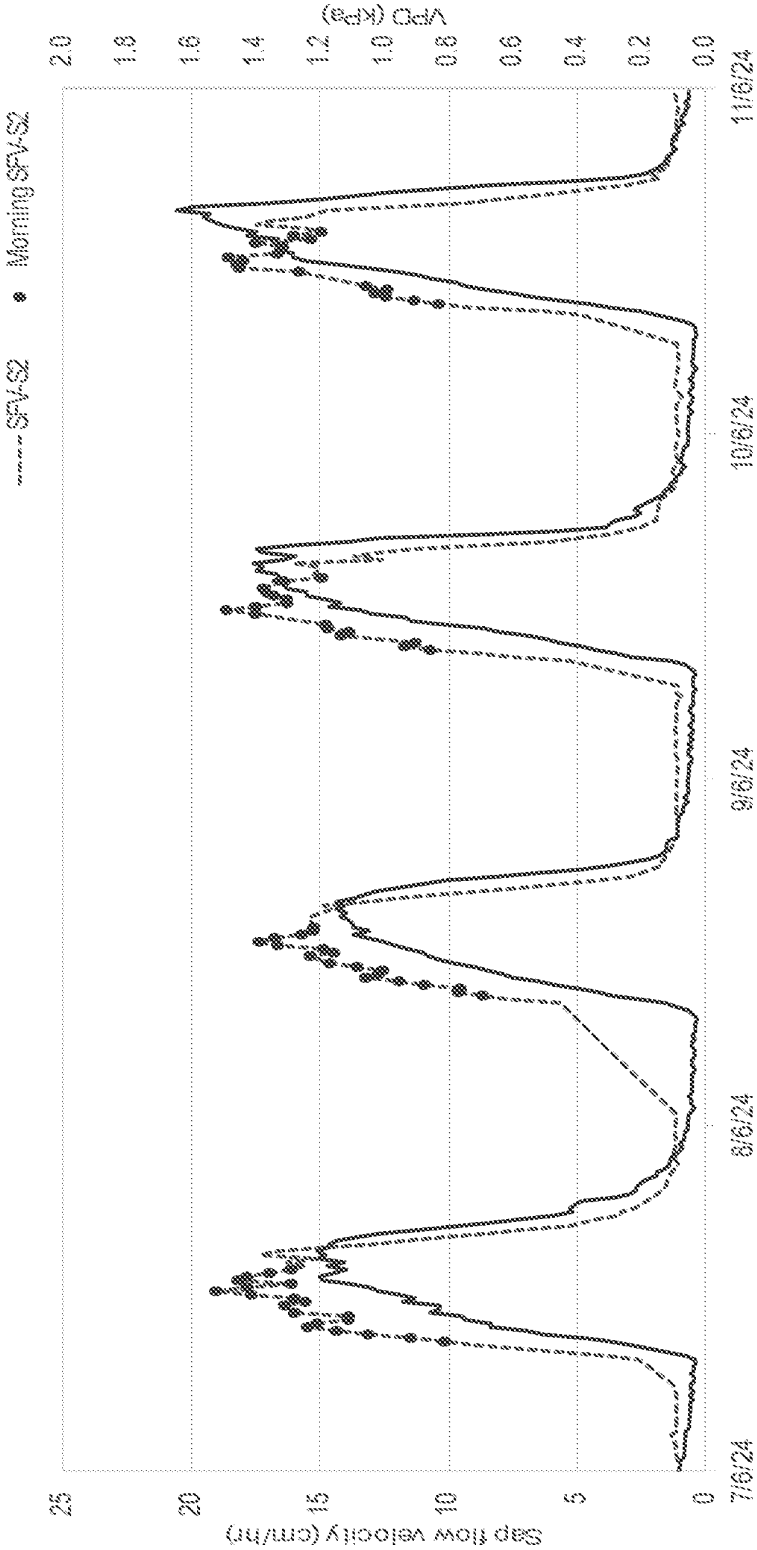


Fig. 15

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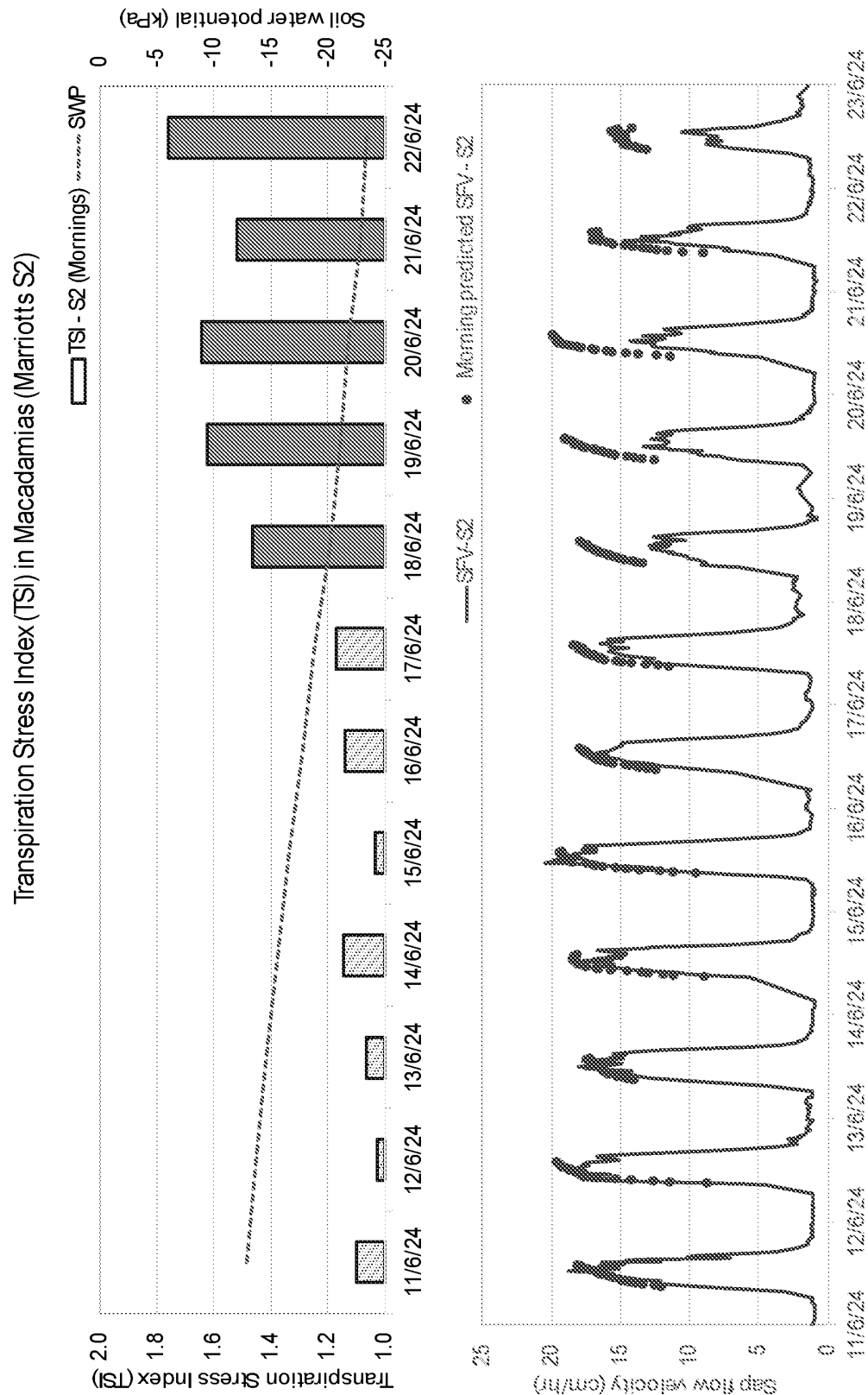


Fig. 16

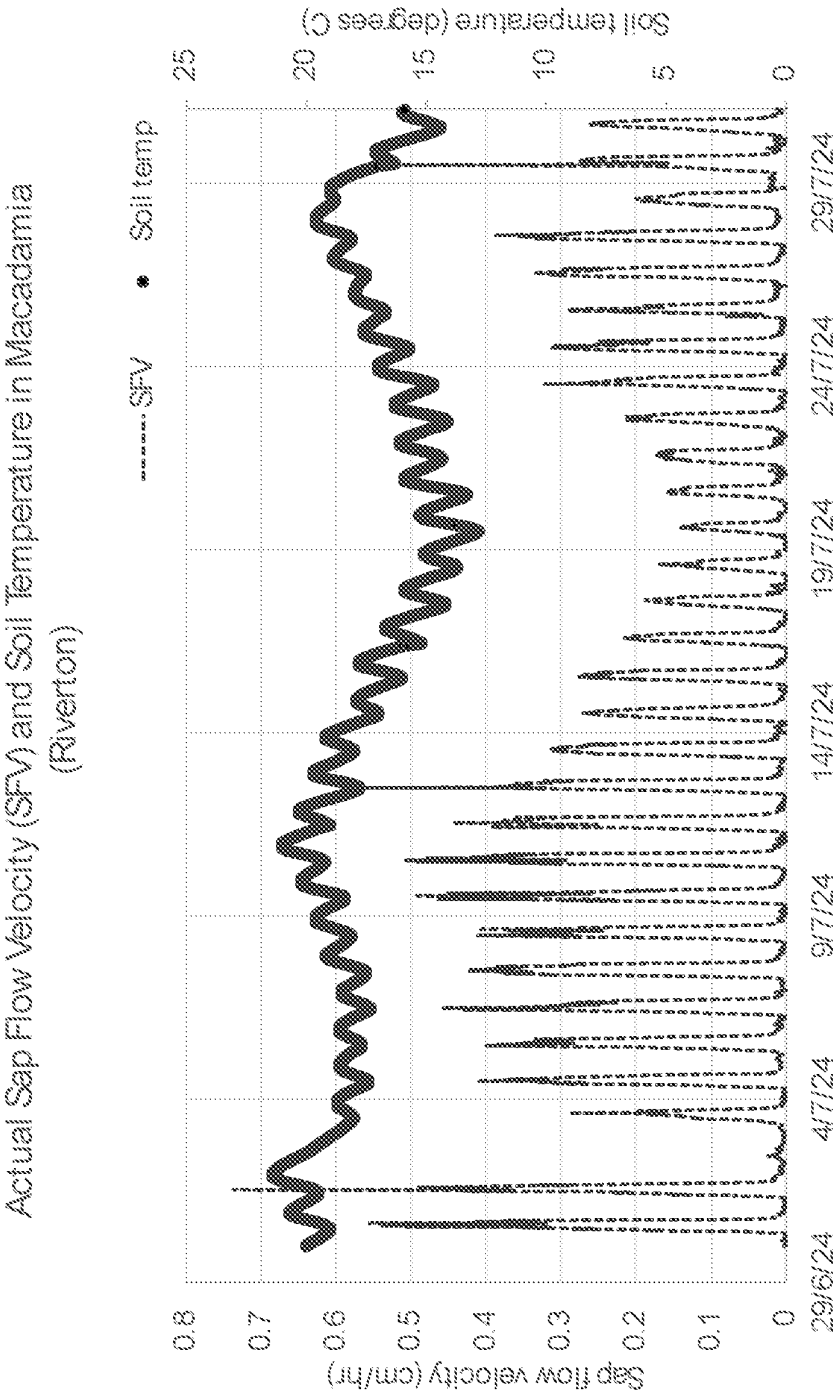


Fig. 17



Fig. 18

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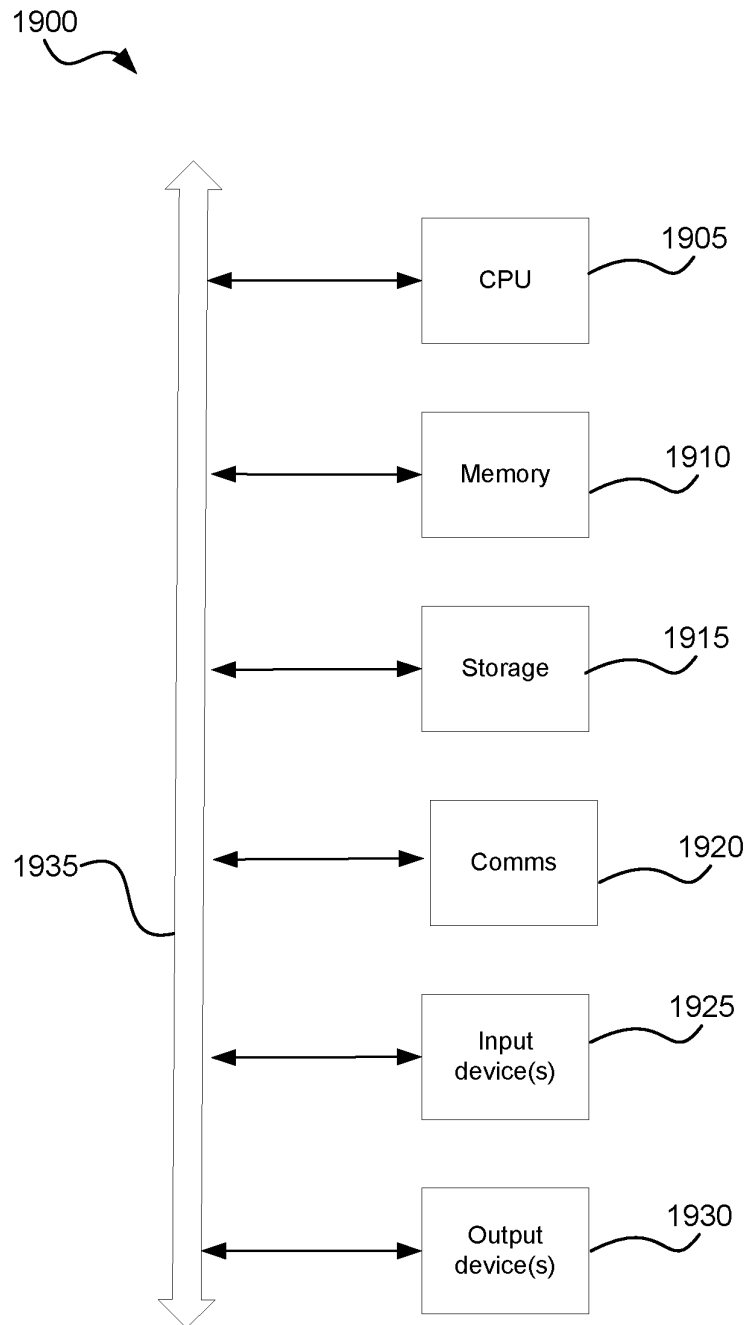


Fig. 19

INTERNATIONAL SEARCH REPORT

International application No.

PCT/AU2024/050834

A. CLASSIFICATION OF SUBJECT MATTER

G01N 33/00 (2006.01) A01G 25/16 (2006.01) G01N 33/24 (2006.01) G06Q 50/02 (2024.01)

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

Google, google scholar and google patents: Keywords: Transpiration/Crop/plant water/drought stress/pressure/force/ index/rate/gradient – sap/stem/stalk flow/velocity – correlation/mathematic/statistical/fitting/predictive function/relation/model/parameter/coefficient/vapor pressure deficit/soil temperature/soil moisture/ solar radiation/light/soil water potential/ humidity/temperature/wind speed – irrigation control/management. PATENW: IPC/CPC: G01N33/0098 or G06Q50/02 or A02G25/16; Keywords: Transpiration/Crop/plant water/drought stress/pressure/force/ index/rate/gradient – sap/stem/stalk flow/velocity – correlation/mathematic/statistical/fitting/predictive function/relation/model/parameter/coefficient/vapor pressure deficit/soil temperature/soil moisture/ solar radiation/light/soil water potential/ humidity/temperature/wind speed – irrigation control/management

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
	Documents are listed in the continuation of Box C	

☒ Further documents are listed in the continuation of Box C☒ See patent family annex

* Special categories of cited documents:	
"A" document defining the general state of the art which is not considered to be of particular relevance	"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"D" document cited by the applicant in the international application	"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"E" earlier application or patent but published on or after the international filing date	"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)	"&" document member of the same patent family
"O" document referring to an oral disclosure, use, exhibition or other means	
"P" document published prior to the international filing date but later than the priority date claimed	

Date of the actual completion of the international search
1 November 2024

Date of mailing of the international search report
01 November 2024

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INTERNATIONAL SEARCH REPORT		International application No.
C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		PCT/AU2024/050834
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2005121536 A1 (BAVEL MICHAEL V) 09 June 2005 Abstract, Fig. 1, 5, 7 Para [0003]-[0005], [0017], [0019], [0056], [0058]-[0060], [0076], [0079], [0132].	1-49
X	CN 113141933 A (GANSU DESERT CONTROL RES INST) 23 July 2021 Abstract, Para [0052], [0060], [0087]-[0088], [0095],[0121], [0123], [0125].	1-49
A	CN 114529102 A (FUZHOU CITY PLANNING AND DESIGN RES INSTITUTE GROUP LIMITED COMPAN) 24 May 2022 Whole document	1-49
A	Su, Y., Wang, X., Sun, Y., & Wu, H. (2022). Sap flow velocity in <i>Fraxinus pennsylvanica</i> in response to water stress and microclimatic variables. <i>Frontiers in Plant Science</i> , 13, 884526, Published in 10 May 2022. DOI: https://doi.org/10.3389/fpls.2022.884526 Whole document	1-49

Form PCT/ISA/210 (fifth sheet) (July 2019)

Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:
the subject matter listed in Rule 39 on which, under Article 17(2)(a)(i), an international search is not required to be carried out, including
2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:
3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a)

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

See Supplemental Box for Details

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.
2. ☒ As all searchable claims could be searched without effort justifying additional fees, this Authority did not invite payment of additional fees.
3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:
4. ☐ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:

Remark on Protest

- ☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.
- ☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.
- ☐ No protest accompanied the payment of additional search fees.

Supplemental Box**Continuation of: Box III**

This International Application does not comply with the requirements of unity of invention because it does not relate to one invention or to a group of inventions so linked as to form a single general inventive concept.

This Authority has found that there are different inventions based on the following features that separate the claims into distinct groups:

- Group I: Claims 1-7, 22-28, 43, 47 are directed to a method/system for detecting transpiration stress for at least one plant and calculating the transpiration stress index. The feature of receiving a plurality of predicted Sap Flow Velocity (SFV) data values associated to the at least one plant and receiving a plurality of predicted SFV data values associated to the at least one plant and calculating the transpiration stress index (TSI) from the above data values and automatically controlling an irrigation system at least partly based on the calculated TSI is specific to this group of claims.
- Group II: Claims 8-14, 29-35, 45, 48 are directed to a method/system for calculating predicted Sap Flow Velocity (SFV) for at least one plant. The feature of receiving a plurality of environmental data values associated to the at least one environmental variable, determine a predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values is specific to this group of claims.
- Group III: Claims 15-21, 36-42, 44, 46, 49 are directed to a method/system for determining correlation function for calculating predicted Sap Flow Velocity for at least one plant. The feature of receiving a plurality of environmental data values associated to the at least one environmental variable, determine a predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values is specific to this group of claims.
- Group IV: Claim 44 is directed to a transpiration stress detection system. The feature of receiving a plurality of predicted Sap Flow Velocity (SFV) data values associated to the at least one plant and receiving a plurality of environmental data values associated to the at least one environmental variable, determine a plurality of predicted SFV data values for the at least one plant using a correlation function configured to correlate measured Sap Flow Velocity (SFV) associated to the at least one plant and the at least one environmental variable, the correlation function configured to take as input the plurality of environmental data values, and calculate a Transpiration Stress Index (TSI) at least partly from the measured SFV data values and the predicted SFV data values is specific to this group of claims.

PCT Rule 13.2, first sentence, states that unity of invention is only fulfilled when there is a technical relationship among the claimed inventions involving one or more of the same or corresponding special technical features. PCT Rule 13.2, second sentence, defines a special technical feature as a feature which makes a contribution over the prior art.

When there is no special technical feature common to all the claimed inventions there is no unity of invention.

In the above groups of claims, the identified features may have the potential to make a contribution over the prior art but are not common to all the claimed inventions and therefore cannot provide the required technical relationship. Therefore there is no special technical feature common to all the claimed inventions and the requirements for unity of invention are consequently not satisfied *a priori*.

INTERNATIONAL SEARCH REPORT Information on patent family members		International application No. PCT/AU2024/050834	
This Annex lists known patent family members relating to the patent documents cited in the above-mentioned international search report. The Australian Patent Office is in no way liable for these particulars which are merely given for the purpose of information.			
Patent Document/s Cited in Search Report		Patent Family Member/s	
Publication Number	Publication Date	Publication Number	Publication Date
US 2005121536 A1	09 June 2005	US 2005121536 A1	09 Jun 2005
		US 7280892 B2	09 Oct 2007
CN 113141933 A	23 July 2021	CN 113141933 A	23 Jul 2021
CN 114529102 A	24 May 2022	CN 114529102 A	24 May 2022
End of Annex			
Due to data integration issues this family listing may not include 10 digit Australian applications filed since May 2001. Form PCT/ISA/210 (Family Annex)(July 2019)			